

Seeing the Economy through Colored Glasses: Partisanship in Macro and (not in) Micro Expectations*

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Abstract

Households view the macroeconomy through partisan lenses, yet expectations about their own finances remain anchored in personal experience. Using survey data spanning 12 U.S. presidential elections since 1980, we show that macroeconomic expectations exhibit sharp partisan reversals around electoral turnovers, whereas expectations about personal finances and employment risk do not. This distinction helps explain why sharp swings in sentiment translate into little change in household spending. We develop and estimate a Bayesian model of macroeconomic belief formation with partisan bias, partisan sentiment, and extrapolation from personal to aggregate expectations. Partisan polarization was already substantial in the 1980s and resurged after 2016, yet differences in personal economic experiences remain the dominant source of disagreement in macroeconomic expectations. Consistent with this extrapolation mechanism, households expecting personally beneficial policies also anticipate stronger aggregate economic outcomes. Distinguishing between macroeconomic and personal expectations is therefore essential for interpreting swings in household sentiment.

Keywords: Macroeconomic Expectations, Elections, Partisanship, Political Opinions

JEL Codes: E21, E30, E71

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1 Introduction

Both economic inequality and political polarization have risen over the past decades in many countries, including the United States. This paper focuses on one particular manifestation of the polarization with deep economic roots – that of macroeconomic expectations. We argue that such polarization cannot be studied independently without considering the role of rising economic inequality. The diverse economic outcomes experienced by households are an important contributor to the polarized views of the macroeconomy.

In particular, we build on the recent literature documenting a sharp divergence and partisan shift in macroeconomic expectations surrounding U.S. presidential turnovers.¹ Belief differences translate into divergent economic decisions, such as stock investments (Meeuwis et al., 2022), and have important macroeconomic consequences, including central bank credibility and long-run inflation expectations (Binder et al., 2024; Kuang et al., 2024). Recently, some commentators also link partisan politics to the so-called “vibecession”, the unusually negative residual of the post-pandemic consumer sentiment that could not be explained by the headline macroeconomic statistics.²

We show that a wide range of macroeconomic – what we call “macro” – expectations have exhibited clear partisan patterns across past U.S. presidential elections – since the 1980s, though with varying intensity over time. In contrast, and uniquely in our study, we find that households’ perceptions and expectations about their own financial situations – “micro expectations” – exhibit much smaller or even negligible partisan shifts. This finding strikes us as both intuitive and reassuring: households are likely better informed about their personal economic conditions, and partisan beliefs cannot fully override expectations shaped by their heterogeneous and idiosyncratic realities. We use this pattern to explain why observed partisan switches in consumer sentiment around electoral turnover did not translate into actual spending differences of comparable magnitude, as reported by Mian et al. (2021). The reason is that personal expectations affect consumption/saving decisions more directly than macro expectations do.

When respondents’ political preferences are not directly observed, researchers often rely on geographic information to infer partisanship, as in Mian et al. (2021) and Meeuwis et al. (2022).

¹See Kamdar and Ray (2022), Mian et al. (2021), Meeuwis et al. (2022), Stantcheva (2024), and Binder et al. (2024), etc. Partisanship in consumer sentiment has been a recurring theme in the monitoring reports of the Michigan Survey of Consumer Expectations (University of Michigan, 2022, 2024) and Gallup (Evans, 2025). More strikingly, the partisan switch around political turnovers was observed in real-time the week after the 2024 U.S. election in consumer sentiment measures by Morning Consult, as reported in *New York Times* (Casselman, 2024).

²See Economist (2023, 2024b,a); Burn-Murdoch (2024) for media coverage of this topic and Harris and Sojourner (2024); Bolhuis et al. (2024) for further scholarly analysis. Particularly relevant to our paper is the blog post by Cummings and Mahoney (2023), who attributes the mismatch between sentiment and economic fundamentals in part to asymmetric partisan biases between Democrats and Republicans.

We instead impute individual-level partisan preferences using a richer set of demographic variables in addition to geographic locations cross-walked between the [American National Election Studies \(2021\)](#) (ANES) and two major surveys of household expectations – the aforementioned MSC and the *Survey of Consumer Expectations* (SCE) by the Federal Reserve Bank of New York. For each U.S. presidential election cycle from 1980 to 2024, we use estimated voter profiles from individual-level ANES microdata to infer each respondent’s political leanings in the SCE and MSC, conditional on their demographic characteristics and the relevant election year. Estimating profiles for each election enables us to capture time-varying voter profiles across elections ([Vilbig and England, 2026](#); [Brown et al., 2025](#)). Our empirical strategy enables a systematic evaluation of the role of partisanship in shaping economic expectations across the 12 presidential elections.

We first empirically examine how partisanship affects macroeconomic expectations using a difference-in-differences (DiD) regression design centered around an election cycle. To link individual-level and aggregate outcomes, we focus on pairs of micro- and macroeconomic expectations. One such example of a pair is personal job separation probabilities and the aggregate unemployment rate. We consider three distinct channels through which partisanship may shape expectations. First, *partisan bias* refers to persistent differences in expectation levels between Democrats and Republicans throughout the election cycle. Second, *partisan sentiment* captures the idea that expectations shift depending on which party holds office, with changes in leadership prompting corresponding changes in beliefs. Third, *belief extrapolation* describes how individuals map their personal economic outlooks onto the broader economy, projecting micro-level experiences onto macroeconomic forecasts. We find strong evidence that partisanship influences expectations through all three channels, though the magnitude of these effects varies across micro-macro expectation pairs and election cycles.

We develop a model of macroeconomic belief formation that incorporates all three channels. This model formalizes the concept of “seeing the economy through colored glasses”, that households rely on their personal economic experiences to shape their views of the broader macroeconomy. We find empirical support that microeconomic expectations are typically less subjective than macroeconomic ones, and that one’s macroeconomic views are always associated with their own microeconomic expectations, controlling for other information sources such as news.

However, the mapping from individual experiences to macroeconomic beliefs is influenced by factors such as political preferences and sentiment. Our paper also captures how partisanship can shape households’ macroeconomic beliefs, even though it does not entirely override the grounding effect of personal experience. We show that both partisan sentiment and belief extrapolation could lead to sudden switches in household macroeconomic expectations around

political turnovers, as found in the data. Also, through the lens of this model, we show that disagreement in macroeconomic views arises from both disparities in individual economic conditions and the “colored glasses” that distort and amplify perceptions of the broader economy.

One innovation of our paper is that we augment the belief model with a measurement block that models each survey respondent’s true political preference as a latent type, which our imputation infers imperfectly. This modeling device not only allows us to convey the uncertainty in the belief model estimates due to noisy measures of survey respondents’ true types, but also lets the model-implied belief patterns indirectly reveal the error rates of our imputed measures. We use Bayesian methods to jointly estimate belief parameters and measurement errors. To the best of our knowledge, no one has applied Bayesian estimations to estimating survey expectations in a similar context. We estimate the model for all election cycles since 1980 using the rich panel structure of the SCE and the repeated cross-sections of the MSC.

Several key findings emerge from our estimation. Across a wide range of expectation domains, we find that there exists not only a level difference between macroeconomic expectations by the incumbent and opponent parties’ affiliates, but also a varying strength of the link between one’s microeconomic expectations and macroeconomic views. Both state-dependent forces contribute to a sudden divergence or reversal in expectations around times of electoral turnover.

Furthermore, our analysis highlights the evolving influence of partisan affiliation on U.S. households’ macroeconomic perceptions since the 1980s. Contrary to the popular notion that partisan expectations are a recent development, we find that they were noticeably pronounced during the 1980s and the post-2016 period. However, despite this resurgence, partisan factors collectively account for no more than 10 percent of the cross-sectional variation in macroeconomic disagreement. A larger share of the disagreement in macroeconomic expectations in our sample period stems from the disparity of microeconomic experiences as measured by dispersion in microeconomic expectations. These findings caution against overstating the role of partisanship in shaping macroeconomic expectations. Instead, it emphasizes that microeconomic inequality in experiences is the basis of such distinctive views of the macroeconomy by households.

Why does partisanship lead to divergence in macroeconomic expectations? In the last part of this paper, we attempt to provide one explanation for such “colored glasses”. Despite many possible non-economic motivations, we nevertheless find evidence for the importance of self-interest-based rationales underlying people’s subjectivity in macroeconomic views. Intuitively, households expect their preferred party/candidate to enact policies that benefit them personally, but also extrapolate such personal benefits onto rosier views of the macroeconomy. It is not just the consideration of self-interest alone, e.g. personally benefiting from expected future policies, but also the extrapolation of policy benefits from the individual to the macroeconomic level

(“What is good for me personally, would also be good for the macroeconomy”) that matters for such an explanation.

We establish evidence supporting such a hypothesis using the Public Policy submodule of the SCE, which elicits households’ expectations about future changes in an array of macroeconomic policies and their perceived effects on such policies on their *personal* economic well-being. During the sample period between 2013 and 2023, we find strong evidence for self-interest-based extrapolation in one’s perceptions of the macroeconomy. In particular, when one expects future macroeconomic policies to bring about personal benefits, they not only expect an improvement in personal finances on many fronts but also expect a better outlook for the future macroeconomy across domains. Our results closely align with findings such as [Stantcheva \(2020\)](#), [Ferrario and Stantcheva \(2022\)](#), and [Stantcheva \(2024\)](#) regarding the formation mechanisms of policy views.

Previous Literature

Our paper primarily contributes to the expanding literature that links partisanship to expectations and economic decisions, in the context of stock investments ([Addoum and Kumar, 2016](#); [Meeuwis et al., 2022](#); [Cassidy and Vorsatz, 2021](#)), inflation expectations ([Binder, 2023](#); [Bachmann et al., 2021](#); [Choi et al., 2022](#); [Gillitzer et al., 2021](#); [Jeong et al., 2025](#)), consumer sentiment ([Mian et al., 2021](#)), patenting behavior ([Engelberg et al., 2023](#)), corporate investment ([Rice, 2020](#)), home sales, ([McCartney et al., 2024](#)), bank lending ([Dagostino et al., 2023](#)), and credit ratings ([Kempf and Tsoutsoura, 2021](#)). A body of literature provides evidence that the changes in expectations lead to different household economic decisions, such as consumption spending and portfolio choices.³ We show that in domains where individual economic expectations matter more directly, e.g., consumption, polarized macroeconomic expectations do not necessarily translate to meaningfully different outcomes.

Our paper is perhaps closest to [Kamdar and Ray \(2022\)](#), which finds political sentiment to be a major common factor in individuals’ macroeconomic expectations and decisions. Individuals have more optimistic economic expectations when their preferred political candidate is the incumbent. We share the factor view of household expectations, but unlike [Kamdar and Ray \(2022\)](#), we study the role of personal finance as a belief factor, and emphasize the importance of heterogeneity in individual economic circumstances. We also interpret partisanship as a mechanism that affects the factor loading of personal finance on macroeconomic expectations.

Broadly speaking, we also present additional evidence of the subjectivity of households’

³[Burke and Ozdagli \(2023\)](#); [Crump et al. \(2022\)](#); [Ichiue and Nishiguchi \(2015\)](#); [Dräger and Nghiem \(2021\)](#); [Duca-Radu et al. \(2021\)](#); [Wang \(2023\)](#).

macroeconomic expectations. For instance, [Andre et al. \(2022\)](#) shows that households think of different propagation channels of the shocks, in particular, demand- and supply-side mechanisms. [Bursztyn et al. \(2023\)](#) and [Guillochon \(2022\)](#) focus on politically divided news sources. The mechanisms documented in this paper might also reflect the “motivated belief” ([Brunnermeier and Parker, 2005](#)) and the echo chamber effects ([Cookson et al., 2023](#)). Our finding of the importance of idiosyncratic experience in driving macroeconomic expectations also echoes the findings by [Taubinsky et al. \(2024\)](#); [Bordalo et al. \(2026\)](#), showing individual circumstances, not only those in the economic domain, affect macroeconomic expectations of households. Our joint focus on both micro and macro expectations is also related to the different responsiveness of firms to microeconomic versus macroeconomic news ([Born et al., 2022](#)).

Political scientists have long studied the effects of partisanship in survey responses on beliefs about the economy. ([Bullock and Lenz, 2019](#)) It has long been established that, for example, Republicans are more likely to respond than Democrats that deficits rose during the Clinton administration⁴, while Democrats are more likely to respond that inflation rose under the Reagan administration⁵ ([Bullock et al., 2015](#)). As early as the 1960s, political scientists have been aware of the “role of enduring partisan commitments in shaping attitudes towards political objects” ([Campbell et al. \(1960\)](#), p. 135). Using ANES survey data, [Bartels \(2002\)](#) found partisan biases in subjective evaluations of the George H. W. Bush administration in 1990-1992, and even in more objective economic data such as unemployment and inflation in the 1980-1988 period. More recently, [Prior et al. \(2015\)](#) found that the survey expectations about the economy reflect a mix of factual beliefs and wishful opinions, although [Bullock et al. \(2015\)](#) found that such partisanship “cheerleading” effects are reduced when respondents are given financial payments for factually correct responses.⁶

Our framework and findings also corroborate several studies beyond the economic domain. For instance, [Gaines et al. \(2007\)](#) examines how partisanship influences the interpretation of factual information, leading to different opinions despite similar factual beliefs. It highlights that interpretations, rather than factual beliefs, drive opinions, making policy change signals more likely to come from independents and weak partisans. Compared to these studies, our paper formalizes and estimates a model of belief formation based on survey expectations that align with the long-held idea that partisanship is a lens through which individuals perceive reality. Just like social issues, news events, policies, etc, individuals’ views about future economic conditions can also be influenced by such a factor. Lastly, our finding that personal finance expectations are less subject to partisan bias than macroeconomic expectations echoes the

⁴During the Clinton administration, Fiscal Year 2000 saw the first federal budget surplus in nearly 50 years.

⁵During Reagan’s presidency, CPI inflation fell from 10.3% in 1981 to 4.1% in 1988.

⁶See [Bullock and Lenz \(2019\)](#) for a detailed review of the political science literature of partisanship in survey responses. [Drobot et al. \(2024\)](#) provides evidence of this kind in the survey expectations of inflation.

finding of [Conover et al. \(1987\)](#).

2 Partisan Belief Facts

2.1 Inferring Partisan Affiliation using Individual-level Data

In both expectation surveys used in this paper, individual respondents' partisan affiliations are rarely available. In particular, the SCE does not elicit political affiliation data for respondents and only began in 2013. The MSC includes sporadic data on political preferences starting in 2006 (with brief earlier coverage from 1980-1985) and began collecting this information on a consistent monthly basis only in February 2017.

We first fill the data gap by systematically imputing each SCE/MSR respondent's partisan preferences for 12 election cycles from 1980 to 2024. Our imputation relies on the ANES microdata, which includes voters' political party affiliation, actual voting choices for presidential candidates, and various demographic variables.

For each election year y , we estimate the average profiles of Democratic (D) and Republican (R) voters separately using the following probit model.

$$\Pr(\kappa_{it} = p \mid X_i) = \Phi(X_i' \beta_p^{(y)}), \quad p \in \{D, R\}, \quad t = y \quad (1)$$

where κ_{it} is the person i 's bi-partisan voter choice in year t . (Third-party/independent voters are excluded.) $\Phi(\cdot)$ is the standard normal CDF. X_i is the set of demographic variables observed in both ANES and the household expectation survey. (See Table 1). The coefficients β^y capture the time-varying voter profile.

The estimated coefficients are then used to impute the affiliation of each respondent in MSC/SCE for the 4-year window of the same election— $t \in \{t-1, \dots, t+2\}$. If the SCE/MSR respondent has a higher predicted Democratic value than a predicted Republican value, we impute that SCE/MSR respondent as Democratic, and vice versa.

$$\hat{\kappa}_{i,t} = \begin{cases} D, & \text{if } \widehat{Pr}(\kappa_{i,t} = D \mid X_i) > \widehat{Pr}(\kappa_{i,t} = R \mid X_i), \\ R, & \text{otherwise.} \end{cases} \quad t \in \{t-1, \dots, t+2\} \quad (2)$$

Two assumptions underlying our imputation procedure are worth clarifying. First, by including a richer set of individual-level demographic variables in addition to regional information, we can more accurately predict SCE/MSR respondents' political preferences than by simply using geographical (county/state/Census region) information, as done in previous papers such

Variable	SCE	MSC
Age (and age squared)	✓	✓
Sex (male/female)	✓	✓
Ethnicity dummies	✓	✗
Level of education dummies	✓	✓
Employment status dummies	✓	✓
Marriage status	✓	✓
Spouse’s employment status	✓	✗
3+ years in current residence	✓	✗
Homeownership	✓	✓ (from 1992)
Income bracket dummies (high, middle, low)	✓	✓
State of residence dummies	✓	✗
Census region of residence dummies	✗	✓

Table 1: List of demographic variables used in imputation of individual-level partisanship, in SCE and MSC respondents. Ethnicity dummies: White, Black, Asian, Native American, Hispanic. Level of education dummies: high school or less, high school graduate, some college, associate degree, baccalaureate degree, post-baccalaureate degree. Employment status dummies: working, temporarily laid off, unemployed, retired, permanently disabled, homemaker, student. Census region of residence dummies: Northeast, South, Midwest, West.

as [Mian et al. \(2021\)](#). It is common for people living in close geographic proximity to differ in their partisan preferences.

This is supported by Table A.16 in the Appendix, where we compare the imputation accuracy of our method with that based solely on geographic information, on the sub-sample of MSC where the partisan affiliation was actually reported. Note that the MSC started collecting data on respondents’ political affiliations sporadically during the 2004, 2008, and 2012 election cycles, then more comprehensively starting in the 2016 election cycle. We find that our more comprehensive imputation predicts an average of 56.7% correctly, as compared to 49.7% with only geographical information. By election cycle, this difference is as low as 1.2% in 2012, to 12.3% in 2016.⁷

Second, by estimating the voter profile for each election cycle separately, we allow the political affiliation of a respondent of a given set of characteristics to change over time. Political realignment across election cycles has been a common theme in U.S. political history. Indeed, Tables in Section A.1 of the Appendix report year-specific coefficients of the probit model for each survey. It shows that the average voter profiles did vary substantially over time.

⁷In the Appendix, Table A.17 and A.18 also show that our baseline imputation based on the Probit model yields accuracy levels comparable to two alternative machine learning models— random forest and XGBoost, which are trained on the pooled sample of all elections while allowing for non-linear interactions between demographic variables and election cycle dummies, so that it implicitly captures time-varying voter profiles.

2.2 Partisanship is not in Microeconomic Expectations

As a preliminary demonstration of our results, we conduct an empirical exercise similar to [Mian et al. \(2021\)](#) to estimate partisan differences in expectations about key labor market outcomes, based on estimated individual-level partisan affiliations from the previous section. Figure 1 averages economic expectations in Republican and Democratic households, across the three U.S. presidential election cycles of 2016, 2020, and 2024. Households politically aligned with the winning candidate (i.e. Democratic households after the 2020 election in which Joe Biden, the Democratic candidate, won) had lower expectations of a higher national unemployment rate (Subfigure 1a) than those aligned with the losing candidate (i.e. Republican households during the 2020 election). In contrast, this partisan divergence did not occur in expectations about individually perceived job separation and job finding rates (Subfigures 1b and 1c). We present this visual evidence as preliminary support for the idea that partisan shifts in expectations are more pronounced for macroeconomic indicators than for microeconomic ones.

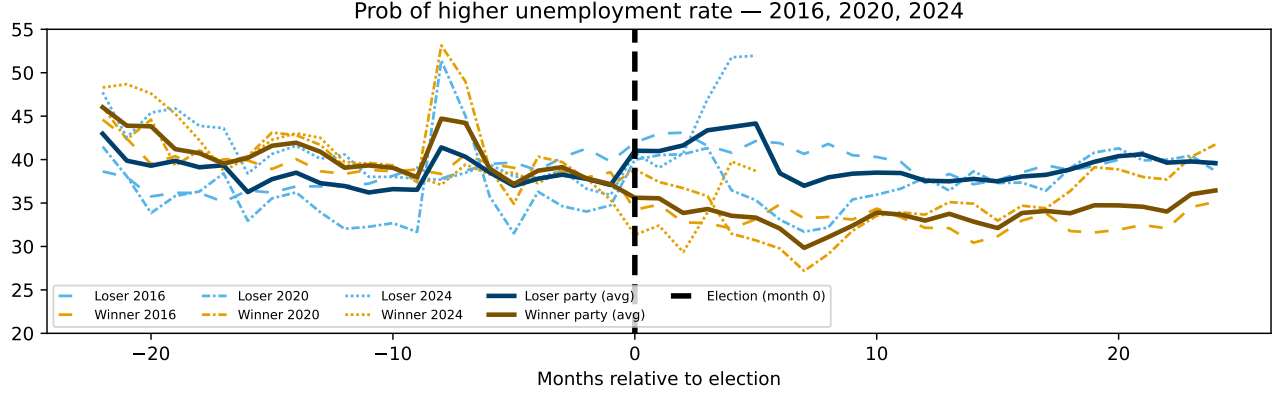
In Figure 2, we show that this extends to general macro- and microeconomic sentiment indicators. We use the same methodology as in Figure 1, but instead with the Index of Consumer Expectations from the University of Michigan Survey of Consumers (MSC). Specifically, we disaggregate the five components of the MSC’s Index of Consumer Expectations into macro- and microeconomic sentiment indicators. We find that our constructed index of macro sentiment (Figure 2a), exhibits a greater partisan switch shortly around elections than our constructed index of micro sentiment does (Figure 2b). Overall, these exercises point towards a larger trend, that partisan voters show greater changes in macroeconomic expectations than in expectations of microeconomic variables or personal finances.

A Historical View. Using the imputed partisan affiliation, we can also examine average expectations of the inferred Democrats and Republicans across twelve presidential elections from 1980 to 2024. In Figure 3, we use MSC microdata of respondents’ macro expectations about the unemployment rate (Figure 3a), macro sentiment (Figure 3b), and micro sentiment (Figure 3c), as well as micro expectations on real household income (Figure 3d) and nominal income (Figure 3e).

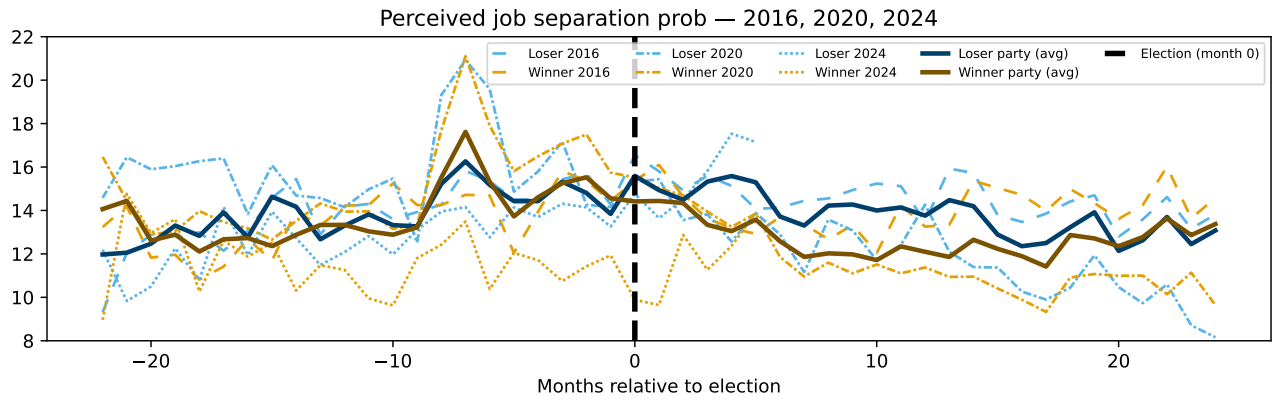
Visually, partisan macro expectations on the unemployment rate and business conditions diverged most around the Republican victories of 1980 and 1984, as well as the more recent 2016 and 2020 elections. In contrast, both parties’ micro expectations of nominal household income have moved in tandem despite persistent level differences and have not exhibited sudden switches across election turnovers. Meanwhile, we can see sudden shifts in patterns of expectations for real income growth, primarily driven by partisan inflation expectations.⁸

⁸Appendix A.4 shows the same pattern with SCE data: the nominal wage and income growth did not exhibit

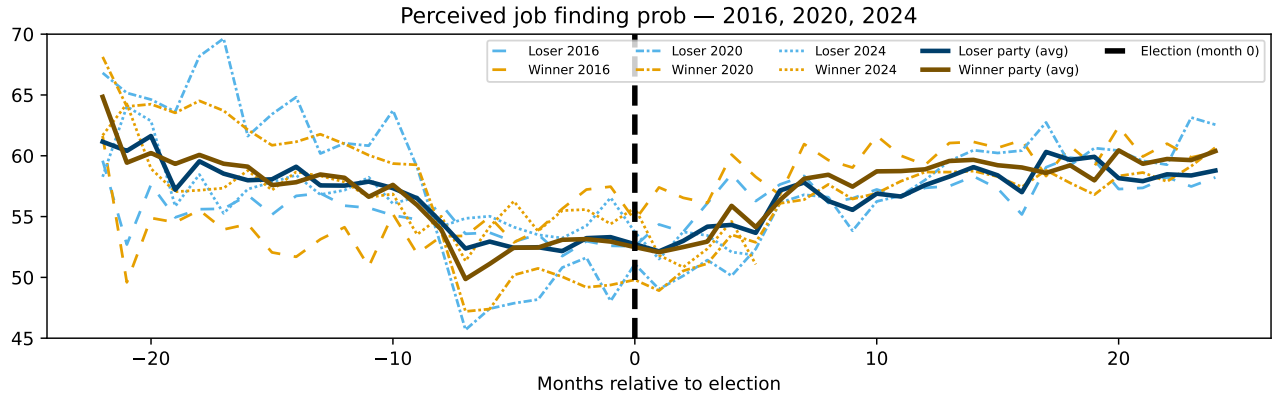
Figure 1: Expectations about Labor Market Outcomes by Individual-level Inferred Partisanship.



(a) Unemployment rate, next 12 months



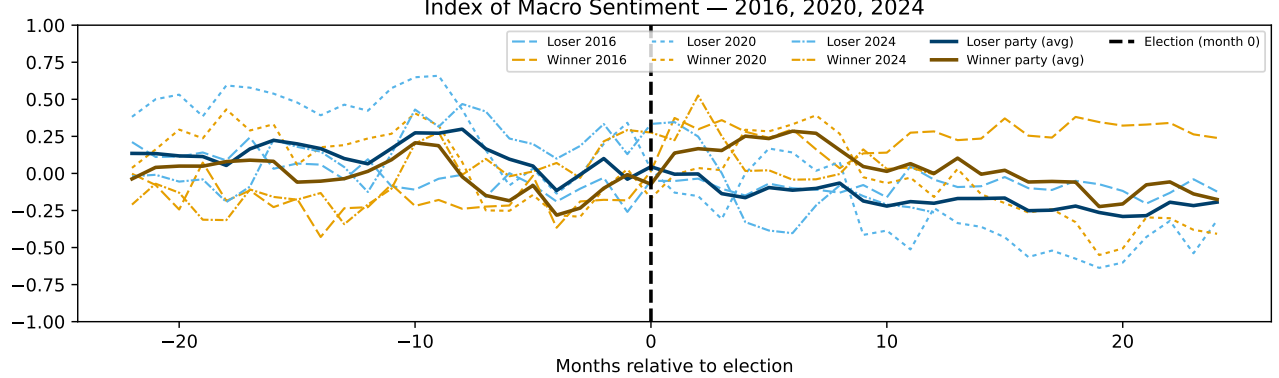
(b) Job separation rate, next 12 months



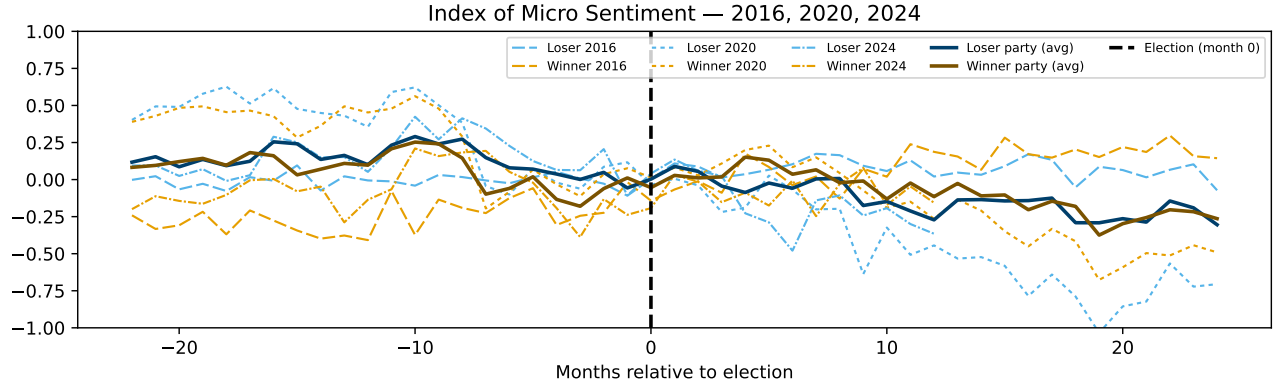
(c) Job finding rate, next 12 months

Notes: Individual political party affiliations are inferred from the procedure in Section 2.1. These tables are generated using data from the 2016, 2020, and 2024 American National Election Study (ANES) and the Survey of Consumer Expectations (SCE) from the calendar year before and after the election. From top to bottom, the figures show mean (inferred) partisan expectations for (i) the perceived probability of a higher nationwide unemployment rate in the next 12 months, (ii) the perceived probability of the respondent being separated from their job in the next 12 months, and (iii) the perceived probability of the respondent finding a new job in the next 3 months conditional on being separated from their job. Orange lines indicate the mean responses of respondents affiliated with the winning party, blue lines indicate those respondents affiliated with the losing one, and the black line indicates the date of the Presidential election.

Figure 2: MSC Macro and Micro Sentiment Indicators by Individual-level Inferred Partisanship.



(a) MSC Macro Sentiment Index



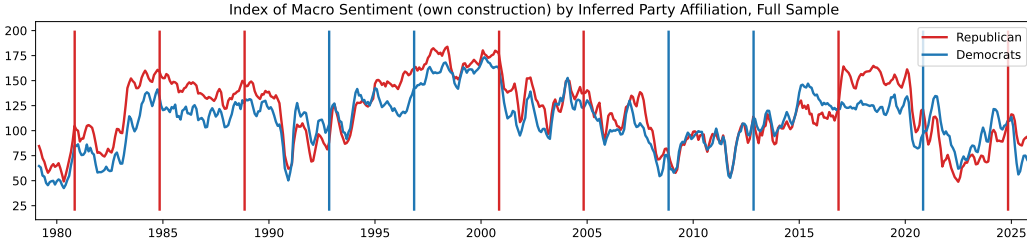
(b) MSC Micro Sentiment Index

Notes: Individual political party affiliations are inferred from the procedure in Section 2.1. These tables are generated using data from the 2016, 2020, and 2024 American National Election Study (ANES) and the Michigan Survey of Consumers (MSC) from the calendar year before and after the election. From top to bottom, the figures show mean (inferred) partisan expectations for (i) the index of macro sentiment includes business conditions in the country as a whole during the next 12 months and 5 years (bus12, bus5), (ii) the micro sentiment includes personal financial situation in past and future 12 months (pago, pexp), good time to buy durable goods (dur). Orange lines indicate the mean responses of respondents affiliated with the winning party, blue lines indicate those respondents affiliated with the losing party, and the black line indicates the date of the Presidential election.

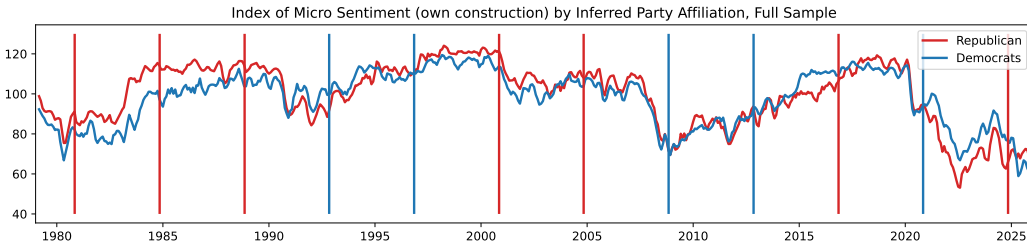
Figure 3: Macro and Micro Expectations by Individual-level Inferred Partisanship, 1980-2024.



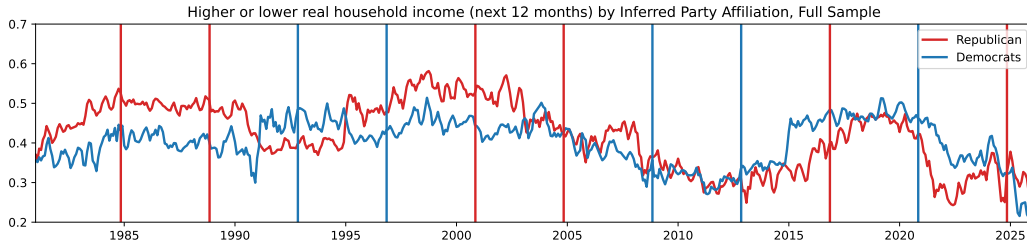
(a) Unemployment rate, 1980-2024



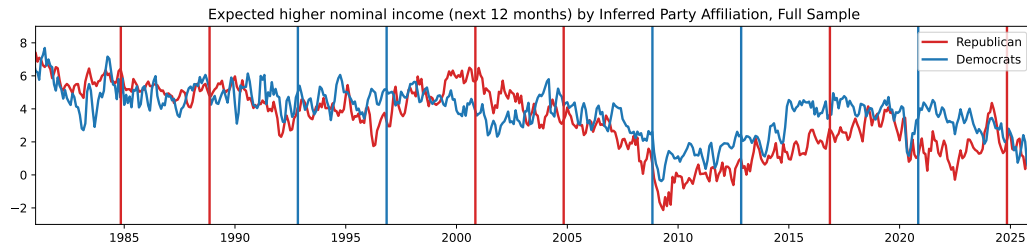
(b) Macro sentiment index, 1980-2024



(c) Micro sentiment index, 1980-2024



(d) Real income, 1980-2024



(e) Nominal income, 1980-2024

Notes: Data from the American National Election Studies and the Michigan Survey of Consumers, January 1979 to November 2025. Red and blue lines show the mean responses of Republicans and Democrats. Vertical lines indicate U.S. presidential election winners by party. All series are 3-month moving averages.

2.3 Consumption Spending Reacts to Micro (not Macro) Expectations

It is natural to ask how the belief patterns reported above explain the actual consumer behaviors. Figure 4 reports two measures of consumption from the household spending module of the SCE: the average change in nominal spending on non-durable consumption over the past 12 months and the share of households purchasing a large durable good within the last four months, grouped by the inferred individual party affiliation measure. The figure suggests no changes in relative spending patterns between Democrats and Republicans around the 2016 or 2020 elections. It seems that the sudden partisan switch in macroeconomic expectations does not translate into meaningful divergence in realized consumption behavior, as also found in [Mian et al. \(2021\)](#).

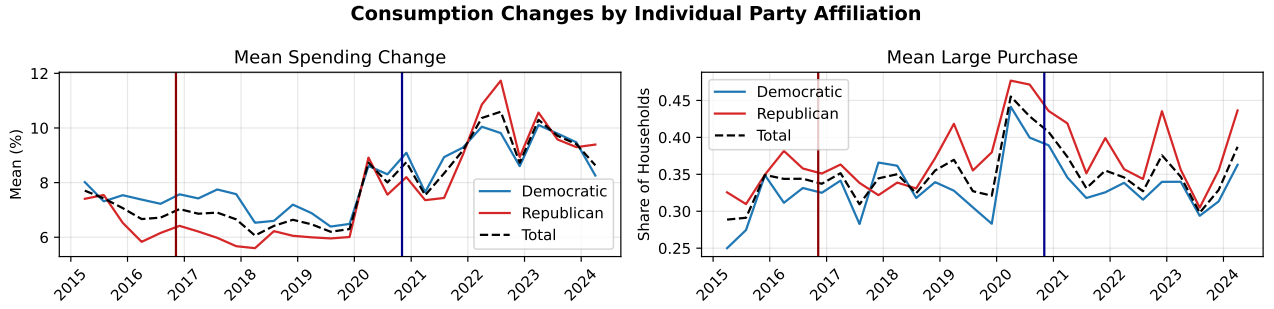


Figure 4: Consumption changes by individual party affiliation

To us, this pattern can be easily reconciled by the fact that households' spending decisions are more directly impacted by expectations about their personal finances than by macroeconomic expectations. We illustrate this point by showing the model-implied consumption responses to empirically estimated shocks to beliefs about unemployment risks. The belief shocks are obtained by estimating changes in each party's beliefs about (i) the aggregate unemployment rate and (ii) individual job separation risk around the 2016 and 2020 election turnovers.⁹ Then we embed these belief shocks in a standard heterogeneous-agent model with idiosyncratic unemployment risks,¹⁰ assuming that households either perceive their idiosyncratic job-loss risk equal to their nationwide unemployment rate beliefs or personal job-separation probabilities.

switches in election turnovers, while their real counterparts did so due to the partisan pattern of inflation expectations, a pattern reported by [Kamdar and Ray \(2022\)](#), [Binder \(2023\)](#), [Kuang et al. \(2024\)](#), etc.

⁹We map these estimates into percentage point changes in perceived unemployment risk by scaling them using the historical distribution of the unemployment rate over the 2016-2024 election cycles. We calculate the differences in post- and pre-election coefficients of the unemployment and job separation probabilities, measured in terms of changes in standard deviation. We then multiply this change by the standard deviation of the unemployment rate within this period to calculate the time paths of the unemployment risk beliefs.

¹⁰The model is quarterly. Households face uninsurable income risks and choose consumption and savings subject to a borrowing constraint. The steady state of the model is calibrated to match an average marginal propensity to consume (MPC) of 0.2. The unemployment spell is assumed to last only one period and is purely transitory.

Figure 5: Unemployment Beliefs over Time

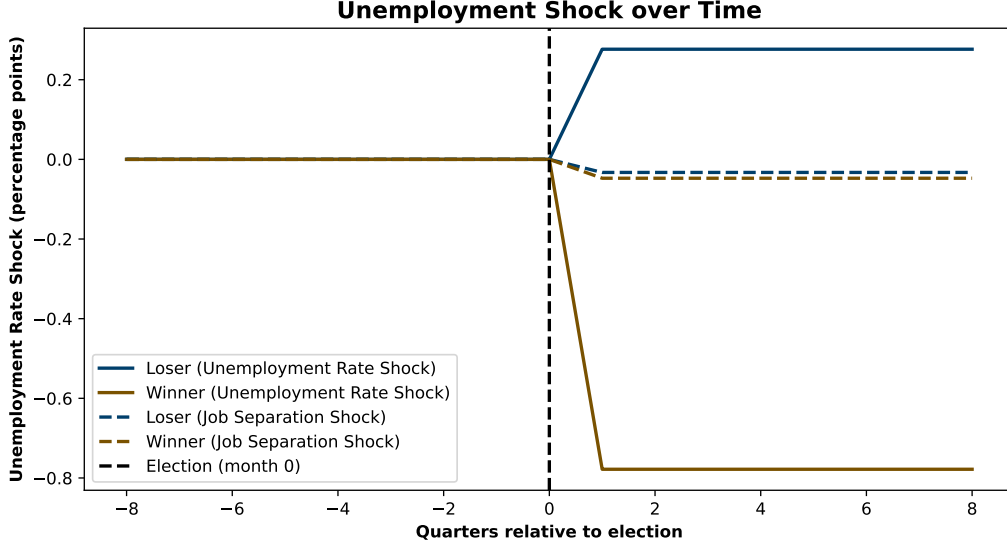


Figure 5 shows that belief divergence is substantially larger for aggregate unemployment than for personal job separation risk.

Figures 6 report the model-implied year-over-year growth rate of quarterly consumption, corresponding to the SCE measure reported above.¹¹ Note that when generating the model-implied consumption response, we isolate the effect of expectations by varying perceived unemployment risk while holding the realized transitions fixed following the approach in Du et al. (2025). In the absence of clear post-election convergence, we model these shocks as persistent for 16 quarters.

Belief shocks based on aggregate unemployment generate large consumption responses that exceed those observed in the survey data. In contrast, shocks based on the risk of personal job separation produce more modest responses that align closely with empirical evidence.

This result suggests that consumption decisions respond primarily to household-specific risk rather than to aggregate macroeconomic expectations. While partisan differences are pronounced for macroeconomic beliefs, they are substantially weaker for individual-level risk, consistent with the limited consumption responses observed in the data.

¹¹In the data, consumption growth is measured as $C^{\text{growth}} = \frac{C_t - C_{t-4}}{C_{t-4}}$, while the model produces deviations from steady state, $\tilde{C}_t = \frac{C_t - C_{ss}}{C_{ss}}$. We therefore recover consumption levels as $C_t = C_{ss}(1 + \tilde{C}_t)$ and construct YoY growth rates accordingly.

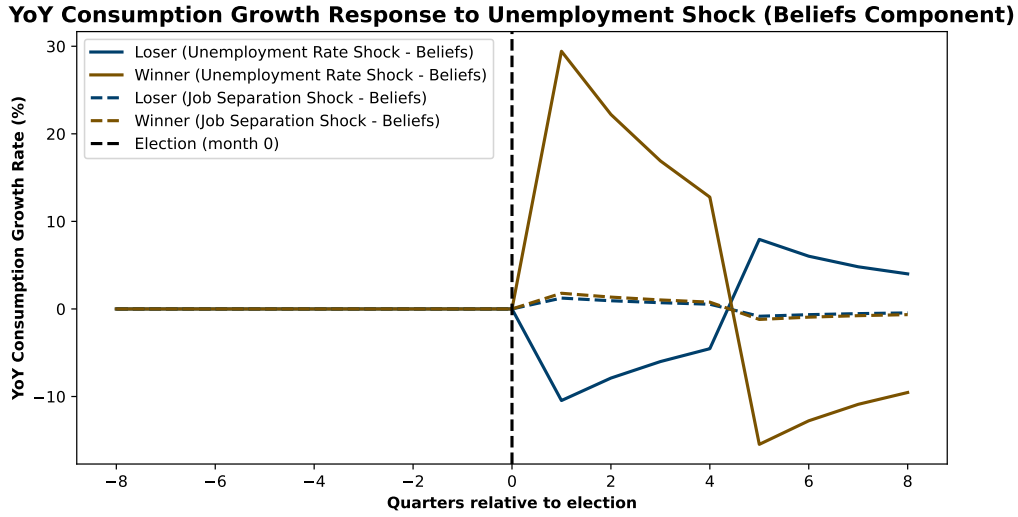


Figure 6: YoY Consumption Growth

3 Regression Analysis of Belief Patterns

3.1 Detecting Partisan Bias and Sentiment

The figures provided above indicate whether there are partisan components to households' expectations before and after the election. Here, we use an event-study-based difference-in-difference regression to formally detect such impacts. We also differentiate between two ways partisanship can influence elicited beliefs. First, *partisan bias* indicates whether Democrats have, on average, different beliefs than Republicans throughout the election cycle. Second, *political sentiment* refers to state-contingent beliefs that an individual might perceive the state of the economy differently depending on whether they are politically aligned with the winner of the presidential election.

We regress elicited beliefs Belief_{it} from SCE or MSC microdata on our imputed party affiliation Dem_i , a dummy variable Win_i if person i rooted for the winning president, a dummy for post-election periods PostElect_t , and an interaction term $\text{Win}_i \times \text{PostElect}_t$:

$$\begin{aligned} \text{Belief}_{it} = & \alpha_0 + \alpha_1 \text{Dem}_i + \alpha_2 \text{Win}_i + \alpha_3 \text{PostElect}_t \\ & + \alpha_4 \text{Win}_i \times \text{PostElect}_t + \text{Controls}_{it} + \varepsilon_{it} \end{aligned} \quad (3)$$

Here, Dem_i is the inferred probability of the respondent being a Democrat supporter. Win_i indicates whether respondent i was affiliated with the party whose candidate won the Presidential election; for example, a Democrat would be designated $\text{Win}_i = 1$ in the 2020 election cycle (when the Democratic candidate Joe Biden won), and 0 in 2024 (when the Republican candidate Donald Trump won).

We interpret the coefficients as follows: $\alpha_1 \neq 0$ indicates partisan *bias*, as households supporting different parties have, on average, a statistically significant difference in expectations of that variable. $\alpha_2 \neq 0$ shows the level difference of respondents who root for the winning party which allows us to pool multiple election cycles together. $\alpha_3 \neq 0$ indicates a post-election shift of all respondents, regardless of partisan affiliation. $\alpha_4 \neq 0$ implies a *partisan sentiment*, as the post-election changes in expectations of households who support the winner change of that election cycle. We control for all household characteristics we used in the imputation stage and cluster standard errors at the individual level. Tables A.19 - A.20 of Appendix A contain estimates for the coefficients of partisan bias and sentiment (α_1 and α_4 respectively) for all micro and macro variables in SCE and MSC for all elections from 1980 to 2000, and from 2004 to 2024 respectively.¹²

In line with the figures, macro expectations tend to be more prone to partisan bias and switches after election results, in that there are more statistically significant coefficients for macro variables than for micro variables. For instance focusing on the most recent elections, the expected probability that the unemployment rate increases (SCE) dropped by 7.4 and 11.2 percentage points (pp) for Republicans after the 2016 and 2024 election and by 7.7 pp for Democrats after the 2020 election. Contrarily, the perceived probability of job separation is only at the 10 percent level statistically significant for the 2024 election. All these effects are not only statistically significant, but also economically meaningful. For the expected inflation rate, Republicans reduced their estimate on average by 0.4 pp in 2016 and 1.4 pp in 2024, while Democrats reduced it by 1.3 pp in 2020.

In conclusion, the statistical analysis confirmed the partisan *bias* and *sentiment* displayed in the figures for macroeconomic variables such as the unemployment or inflation rate. These effects are weaker for microeconomic variables.

3.2 Micro to Macro: Partisan Bias, Sentiment, and Belief Extrapolation

From the analysis in the previous section, we found that households' expectations about their own economic situation (micro beliefs) are less affected by partisanship than expectations about the macroeconomy, e.g. about everyone else.

In both the SCE and MSC surveys, we select pairs of related micro- and macroeconomic expectations. Micro beliefs capture respondents own economic situations, while macro beliefs reflect expectations about the aggregate economy.

We may hypothesize that agents extrapolate from and make statements about the macroeconomy based on their beliefs of their own individual situations. Under that possibility, agents

¹²In these tables, coefficients that are statistically insignificant at the 10% level are displayed as 0.

who are positive in their individual situations (micro beliefs) may think more positive macro beliefs as well.¹³ We can include partisanship into the realm of individual situations; households may see a stronger connection between their own situations and the situation of all other agents (micro to macro) depending on their political preferences or the political affiliation of the incumbent president.

Therefore, in this section we test the hypothesis that households' micro beliefs may also affect macro beliefs, in addition to their political preferences and the political affiliation of the current incumbent. Specifically, we augment the specification in Equation 3 with respondents' micro beliefs:

$$\begin{aligned} \text{MacroBelief}_{it} = & \alpha_0 \\ & + \alpha_1 \text{Dem}_i + \alpha_2 \text{Win}_i + \alpha_3 \text{PostElect}_t + \alpha_4 \text{Win}_i \times \text{PostElect}_t \\ & + \alpha_5 \text{MicroBelief}_{it} + \text{Controls}_{it} + \text{ElectionCycle} + \varepsilon_{it} \end{aligned} \quad (4)$$

where, again, Win_i is an indicator variable equal to 1 if the imputed political affiliation of the respondent is the same as that of the winning candidate in a particular election cycle. Here, MicroBelief_{it} and MacroBelief_{it} are pairs of respondent i 's expectations on micro- and macroeconomic variables.

In this specification, these coefficients have the following interpretations:

- α_1 : Partisan bias of macro variable
- α_4 : Partisan sentiment of macro variable
- α_5 : Partisan bias of extrapolation from micro beliefs to macro beliefs

Similar to Equation 3, α_1 and α_4 capture how partisanship directly affects the expectation of macro beliefs through partisan *bias* and state-contingent *sentiment*, respectively. Additionally, α_5 shows how the extrapolation from micro variables to macro beliefs.

As for what constitutes the variables MicroBelief_{it} and MacroBelief_{it} , we focus on two sets of variables. First, there are the variables on the expected inflation rate of categories of items for which we have survey data, such as gas, food, and rent prices of households. Second, there are the variables regarding the overall economy. For example, the MSC Index of Consumer Sentiment (ICS) is computed from responses of five questions, of which three focus on the

¹³Colloquially, households who are doing well under the current economy may expect that other households are also doing well.

personal situation¹⁴ and two on the macroeconomy.¹⁵ Additionally, also from the MSC data, we can relate expected personal finances (PEXP) with expected conditions of businesses (BEXP).

As for the SCE survey data, we relate the micro beliefs of personal job separation and job finding probabilities, during the next 12 months, with the macro beliefs of the aggregate unemployment rate. Additionally, we use the expected rate of inflation and relate it to changes in gas and rent prices.¹⁶

We re-scale all categorical expectations variables to a scale of $[0, 1]$, where larger values indicate more positive sentiment (i.e. 1 for positive, 0 for negative). For example, the MSC codebook includes a question for respondents' expectations of business conditions 12 months from now (**bus12**). For that question, the categorical responses reach from 1 for good (positive), 3 for neutral, and to 5 for bad (negative). In that case, we adjust the responses to a $[0, 1]$ scale such that 1 indicates positive expectations, while 0 indicates negative expectations, and 0.5 for neutral expectations. For expectation variables with numerical percentage responses, we maintain percentages.

Table 2 summarizes the results pooled across recent election cycles in which the incumbent did not win reelection (2016, 2020, and 2024). A key finding is the strong correlation between micro and macro expectations, captured by coefficient α_5 (Micro): respondents who anticipate increases in personal economic outcomes also tend to expect increases in aggregate outcomes. This correlation varies over time.

In summary, partisanship affects household beliefs about the macroeconomy through channels

¹⁴To construct our measure of micro sentiment from MSC microdata, we use 3 variables:

- Q1 (PAGO) We are interested in how people are getting along financially these days. Would you say that you (and your family living there) are better off or worse off financially than you were a year ago?
- Q3 (PEXP) Now looking ahead—do you think that a year from now you (and your family living there) will be better off financially, worse off financially, or just about the same as now?
- Q19 (DUR) Think about the big things people buy for their homes – such as furniture, a refrigerator, stove, television, and things like that. Generally speaking, do you think now is a good or a bad time for people to buy major household items?

¹⁵To construct our measure of macro sentiment from MSC microdata, we use 2 variables:

- Q5 (BUS12) Now turning to business conditions in the country as a whole—do you think that during the next 12 months we'll have good times financially or bad times financially?
- Q9 (BUS5) Looking ahead, which would you say is more likely – that in the country as a whole we'll have continuous good times during the next 5 years or so, or that we will have periods of widespread unemployment or depression?

The index is created by using the sum of the two variables, dividing by the value of the index in the base year (2003 January), and adding 2.

¹⁶We have found it difficult to find a clear micro-macro pair of expectations for inflation. The SCE also considers prices for items such as food, medical bills, and education, although none of those are specific to all expenditures paid by households.

Table 2: Micro to Macro Partisan Bias and Switch for 2016-2024 Elections

Dep Var Micro Var	MSC Macro Index	MSC Macro Index Micro Index	MSC Bexp	MSC Bexp Pexp	SCE Unemp	SCE Unemp Job Sep	SCE Unemp Job Find	SCE Inflation	SCE Inflation Gas Price	SCE Inflation Rent
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Dem	-5.500** (2.431)	-7.859*** (2.649)	-0.033 (0.022)	-0.040* (0.023)	0.937 (1.704)	1.249 (1.888)	1.047 (2.090)	-0.514*** (0.176)	-0.345 (0.303)	-0.364 (0.256)
WinRoot	-16.360*** (2.534)	-13.698*** (1.932)	-0.051** (0.021)	-0.043*** (0.013)	-1.450** (0.600)	-1.138 (0.764)	-1.225* (0.693)	-0.270 (0.285)	-0.176 (0.263)	-0.168 (0.185)
postElect	-27.817* (16.538)	-16.129*** (5.147)	-0.126* (0.069)	-0.091** (0.038)	5.967*** (0.431)	5.311*** (0.398)	5.961*** (0.442)	1.182*** (0.266)	1.468*** (0.234)	0.997*** (0.155)
WinRoot $\times$ postElect	31.944*** (4.873)	27.856*** (3.715)	0.158*** (0.037)	0.130*** (0.022)	-8.008*** (1.086)	-8.624*** (1.136)	-9.370*** (1.063)	-3.095*** (0.725)	-1.955*** (0.423)	-1.663*** (0.403)
Micro		0.941*** (0.028)		0.428*** (0.029)		0.242*** (0.013)	-0.013 (0.011)		0.207*** (0.046)	0.319*** (0.016)
Controls	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Adj. R^2	0.063	0.239	0.039	0.172	0.021	0.060	0.022	0.076	0.130	0.165
N	51949	51949	51949	51949	130924	79506	79528	128334	110456	110446

beyond bias and sentiment. The degree to which changes in microeconomic circumstances get extrapolated into macroeconomic variables depends significantly on partisanship.

3.3 Controlling for the Role of Macroeconomic News

Households are not only restricted to form their macro expectations using their individual situation, but receive *news* from media outlets (TV, internet, newspapers, etc.), or colleagues and friends. Being exposed to additional information about the macro economy, such as positive/negative news about the economy, employment, or inflation, will influence how they predict the future.

Therefore, we extend the specification in Equation 4 with a *News* indicator which equals 1 if the respondent received positive news, 0 if the respondent receives no news, and -1 if the respondent received unfavorable news within the last few months.

$$\begin{aligned} \text{MacroBelief}_{it} = & \alpha_0 + \alpha_1 \text{Dem}_i + \alpha_2 \text{Win}_i + \alpha_3 \text{PostElect}_t + \alpha_4 \text{Win}_i \times \text{PostElect}_t \\ & + \alpha_4 \text{MicroBelief}_{it} + \alpha_5 \text{News}_{it} + \text{Controls}_{it} + \text{ElectionCycle} + \varepsilon_{it} \end{aligned} \quad (5)$$

Using the MSC, we can directly observe whether households received news about a broad set of economic topics, as well as specific information related to prices and unemployment. This allows us to avoid controlling for aggregate macroeconomic fluctuations or inferring who was exposed to which news since this information is available directly from the MSC data. In Table 3, we use the overall news index to study its relationship with the macro sentiment index (columns 1-2), employment-related news to analyze expectations about the unemployment rate (columns 3-4), and price-related news to examine inflation expectations (columns 5-7).

First, the coefficient on *News* (α_6) is highly significant across all specifications. This indicates that when agents receive positive news, they revise their expectations about macroeconomic

Table 3: Micro and News to Macro Partisan Bias and Switch for 2016-2024 Elections

Dep Var Micro Var News Var	MSC Macro Index (1)	MSC Macro Index Micro Index All News (2)	MSC Unemp (3)	MSC Unemp Job Separation Employment News (4)	MSC Inflation (5)	MSC Inflation Gas Price Inflation News (6)	MSC Inflation Nominal Income Inflation News (7)
Dem	-5.500** (2.431)	-4.197 (3.731)	0.038** (0.019)	0.039* (0.020)	-0.070 (0.337)	-0.027 (0.392)	-0.018 (0.320)
WinRoot	-16.360*** (2.534)	-10.387*** (1.970)	0.063*** (0.002)	0.056*** (0.002)	0.025 (0.307)	0.049 (0.244)	0.038 (0.240)
postElect	-27.817* (16.538)	-13.614*** (4.390)	0.068 (0.050)	0.059 (0.044)	1.735 (1.093)	1.724** (0.707)	1.541 (0.953)
WinRoot $\times$ postElect	31.944*** (4.873)	21.756*** (4.595)	-0.144*** (0.022)	-0.131*** (0.020)	-0.435 (0.469)	-0.594 (0.442)	-0.469 (0.407)
Micro		0.770*** (0.017)		0.168*** (0.014)		0.029*** (0.004)	-0.031** (0.014)
News		42.944*** (2.328)		-0.142*** (0.025)		-1.362*** (0.280)	-1.485*** (0.120)
Controls	YES	YES	YES	YES	YES	YES	YES
Adj. R^2	0.063	0.309	0.056	0.093	0.063	0.115	0.073
N	51949	51949	51949	51949	51365	29226	47025

variables in a more optimistic direction; for example, expressing more positive macro sentiment or expecting lower unemployment or inflation. Importantly, the earlier finding that microeconomic beliefs influence macroeconomic expectations remains robust. Thus, the paper’s central mechanism holds even after accounting for the role of news.¹⁷

4 An Expectation Model with Partisanship

The previous sections established several different channels through which partisanship affects macroeconomic expectations. To formally capture these mechanisms, we introduce a state-dependent model of macroeconomic expectations that can be flexibly estimated from survey data. Importantly, the model estimation also treats the empirically imputed political leaning of each survey respondent as a noisy signal of their true type.

4.1 Setup

We begin by considering a simple benchmark environment in which partisanship does not influence macroeconomic expectations. Each agent i observes their own idiosyncratic (micro-level) state $s_{i,t}$ with high precision, but imperfectly observes the state of the aggregate economy, denoted by ψ_t . Agents form subjective expectations about the aggregate state, which are

¹⁷Related to our estimates, Jeong et al. (2025) studies how the same self-reported news exposures affects MSC households’ inflation expectations differently among incumbent versus opposition party’s affiliates. We show that conditional on the news’s impacts, personal finance expectations remain important drivers of macroeconomic expectations.

recorded in survey data as $\tilde{\psi}_{i,t}$. These expectations are formed in part as a function of the agent's individual circumstance, which is governed by the mapping $z(\cdot)$.

In addition, expectations are affected by common shocks ϕ_t that are shared across all agents, and by idiosyncratic measurement errors $\epsilon_{i,t}$.¹⁸ The common component ϕ_t captures aggregate sources of variation in expectations across respondents, such as changes in macroeconomic fundamentals, common information updates, shifts in sentiment, and related factors.

$$\tilde{\psi}_{i,t} = z(s_{i,t}) + \phi_t + \epsilon_{i,t} \quad (6)$$

We assume the form of $z(\cdot)$ to be a linear function of $s_{i,t}$,

$$z(s_{i,t}) = \alpha + \lambda s_{i,t}, \quad (7)$$

where λ measures the sensitivity of the macroeconomic expectations with respect to $s_{i,t}$. One interpretation of such a functional form is that $s_{i,t}$ is a noisy signal of ψ_t , and α is some prior belief about the aggregate economy $\psi_{i,t}$. $z(s_{i,t})$ is formed based on non-sophisticated Bayesian updating, in which agents do not necessarily optimally decide λ based on information precision. $z(\cdot)$ can also be interpreted as an ‘‘attribution’’ function, in that it measures how much individuals attribute their future circumstances to expected macroeconomic conditions. Broadly, one can view $s_{i,t}$ as a belief factor driven by personal experiences that loads onto macro beliefs.

In order to capture partisanship-induced belief differences and admit other unspecified sources of variations of beliefs across individuals and time, we assume instead that $z(\cdot)$ is state-dependent, varying with two other variables $\kappa_{i,t}$ and $x_{i,t}$, as specified below.

$$\begin{aligned} z(s_{i,t}, x_{i,t}, \kappa_{i,t}) &= \mathbf{1}(\kappa_{i,t} = D)\omega + \mathbf{1}(x_{i,t} = 1)z^{up}(s_{i,t}) + \mathbf{1}(x_{i,t} \neq 1)z^{down}(s_{i,t}) \\ x_{i,t} &= \mathbf{1}(\kappa_{i,t} = D)\mathbf{1}(\Theta_t = D) + \mathbf{1}(\kappa_{i,t} = R)\mathbf{1}(\Theta_t = R) \\ z^{up} &= \alpha^{up} + \lambda^{up}s_{i,t} \\ z^{down} &= \alpha^{down} + \lambda^{down}s_{i,t} \end{aligned} \quad (8)$$

Here, $\kappa_{i,t}$ captures the partisan affiliation of agent i at time t . For simplicity, we assume $\kappa_{i,t}$ is either one of D or R , but this can be generalized to more than two parties. Therefore, ω is meant to capture the mean difference in the levels of macroeconomic expectations held by D compared to that by R , i.e. partisan bias. We assume this difference arises from strong priors

¹⁸In principle, there also exist individual fixed effects that are time-invariant. We do not model these explicitly to avoid confounding them with the effects operating through $\kappa_{i,t}$, which is theoretically time-varying but empirically constant at the individual level over a short panel history.

due to partisanship.

The second part of the beliefs, z^{up} and z^{down} , captures partisan sentiment and extrapolations that vary with the current political landscape. $x_{i,t}$ depends not only on one’s political preference but also on whether their party is the incumbent. $\mathbf{1}(x_{i,t})$ equals 1 when agent i ’s preferred candidate/party is in power. Depending on its value, z^{up} or z^{down} , either one of the state-dependent belief functions is triggered, as the agent’s belief mapping from their micro condition to the macroeconomy. Although not imposed as a restriction, in general, it may be natural to assume that both $z^{up}(\cdot)$ and $z^{down}(\cdot)$ share signs but have different slopes ($\lambda^{up} \neq \lambda^{down}$), indicating that the responsiveness of macroeconomic belief to idiosyncratic signals is state-dependent. Furthermore, we assume the constant term α also differs across the two states, e.g. $\alpha^{up} \neq \alpha^{down}$, implying that partisanship also induces level shifts in macroeconomic beliefs independent of one’s micro expectations. The difference in sentiment may stem from differing interpretations of commonly available macroeconomic news by incumbent and opposition party affiliates. We refer to the scenario in which λ^{up} is not equal to λ^{down} as partisanship in extrapolation, and the scenario in which α^{up} differs from α^{down} to the partisan shift in the sentiment.

On the one hand, our model makes the intuitive assumption supported by our regression results, that personal economic expectations serve as a basis for one’s macroeconomic expectations. Partisanship does not overrule its relevance in household expectations about the macroeconomy. On the other hand, it captures the role of partisanship as a lens between households’ perceptions of the macroeconomy and their personal economic conditions.¹⁹

With such a formulation, we have a mapping between agent i ’s microeconomic expectations at time t , $s_{i,t}$, to their macroeconomic expectations $\tilde{\psi}_{i,t}$, as a function of their underlying partisanship $\kappa_{i,t} \in \{D, R\}$ and their alignment with the incumbent party, $x_{i,t} \in \{0, 1\}$.

4.2 Model Implications

Our model differentiates three mechanisms which we have broadly called partisanship, as estimated by the reduced form regressions in Section 3.5: (a) ω , the partisan bias unconditional on the current state of the macroeconomy and politics, estimated as α_1 in regressions specified in Equation 3. (b) α^{up} and α^{down} , the post-election sentiment shift in macroeconomic expectations when one’s preferred candidate/party wins or loses the presidency; (c) state-dependent extrapolations, e.g. different λ^{up} and λ^{down} , that lead to varying degrees of sensitivity between individual expectations and macroeconomic expectations. It is easy to show that either (b)

¹⁹It is possible that partisanship is simply one of the many factors, or “mindsets” in the words of [Chinoy et al. \(2023\)](#), that affect how people view information and reality. We leave it for future research to generalize this insight.

or (c) could contribute to a sudden switch in the party-specific macroeconomic expectations around elections, witnessing partisan changes. This mechanism exactly corresponds to the non-zero estimate of α_3 in the reduced form regression as in Equation 3. Such a switch around election outcomes has been a salient pattern in various macroeconomic expectations, as shown in Section 3.

Figure A.3 in Online Appendix B shows simulated results on how either non-identical α s or non-identical λ s can result in a partisan switch in macroeconomic expectations around an election turnover (a change in the incumbent party). Meanwhile, a non-zero ω implies partisan bias between two parties, and it alone does not cause a switch in expectations around elections with political turnover. Meanwhile, the microeconomic expectations $s_{i,t}$ are not subject to such partisan switches, consistent with our empirical findings. Note that we also take the distribution and dynamics of $s_{i,t}$ as given. It may be the case that there are constant differences in microeconomic expectations along partisan lines, reflecting persistent differences between the two parties' supporters in their economic conditions.

The exact degree of importance of each of these channels can be uncovered by estimating such a model using rich microdata spanning multiple election cycles. We can also empirically investigate if the partisanship in macroeconomic expectations has intensified over time and depends on the specific domain of economic expectations.

4.3 Estimation of the Model

Modeling partisanship labeling errors. A central identification challenge is that an individual's true partisan type $\kappa_{i,t}$ is not directly observed. In the previous section, we inferred it as $\hat{\kappa}_{i,t} \in \{D, R\}$ by simply comparing the imputed partisan probabilities from the probit model, $\widehat{prob}(\kappa_{i,t} = D)$ and $\widehat{prob}(\kappa_{i,t} = R)$, i.e.,

$$\hat{\kappa}_{i,t} = \begin{cases} D, & \text{if } \widehat{prob}(\kappa_{i,t} = D) > \widehat{prob}(\kappa_{i,t} = R), \\ R, & \text{otherwise.} \end{cases} \quad (9)$$

Here, we augment the belief model with a measurement block that explicitly specifies the possibility that the generated $\hat{\kappa}_{i,t}$ are subject to labeling errors.²⁰ In particular, we allow for misclassification between observed type $\hat{\kappa}_{i,t}$ and true type $\kappa_{i,t}$. Let

$$\kappa_{i,t} \sim \text{Bernoulli}(p), \quad (10)$$

²⁰In Appendix C.2, we alternatively estimate the model in the frequentist approach by maximum-likelihood. There, we do not model the mislabeling. Instead, we use the imputed probability (instead of the binary flag) of one's partisan affiliation to estimate the model. The findings are rather similar to the results reported here.

and define false-negative and false-positive rates as

$$e_1 = P(\hat{\kappa}_{i,t} = R \mid \kappa_{i,t} = D), \quad e_0 = P(\hat{\kappa}_{i,t} = D \mid \kappa_{i,t} = R). \quad (11)$$

The former governs the probability of a true Democrat being labeled as a Republican, while the latter governs the fraction of a true Republican labeled as a Democrat.

By explicitly modeling labeling error rates and acknowledging this uncertainty about the generated type flag, the model not only avoids treating the imperfectly imputed political type as the ground truth, but also lets the survey belief patterns provide additional information about the true type of the survey respondent.

For likelihood estimation, we also assume that the residual $\varepsilon_{i,t}$ in observed beliefs (Equation 8) is i.i.d. and normally distributed with mean zero and variance σ^2 . It captures idiosyncratic noise or measurement error specific to individual i and period t .

Bayesian estimation. Incorporating labeling error into the model allows for joint identification of the belief-formation parameters in Equation 8, the population share of Democratic party affiliates in the population, p , and the misclassification rates (e_0, e_1) . The core parameter vector thus consists of $\Gamma = [\lambda^{up}, \lambda^{down}, \alpha^{up}, \alpha^{down}, \omega]$, augmented by (p, e_0, e_1) and the idiosyncratic shock scale σ . Rather than treating $\hat{\kappa}_{i,t}$ as observed without error, the model specification explicitly decomposes the observed variation into components arising from the underlying belief-formation process and from uncertainty in the upstream classification of partisan type.

We estimate the model using Bayesian methods implemented in PyMC (see Appendix C for details). Because the true partisan type $\kappa_{i,t}$ is latent and discrete, the likelihood is analytically marginalized over $k \in \{D, R\}$ at each observation (i, t) , as shown in Equation 12. Specifically, the model evaluates the joint likelihood of the imputed label $\hat{\kappa}_{i,t}$, the macro belief $\tilde{\psi}_{i,t}$, and the micro signal $s_{i,t}$ as a weighted mixture over the two possible true states, combining the misclassification process with a Normal observation model for macroeconomic expectations. This marginalized representation keeps the likelihood continuous in the unknown parameters while preserving the measurement-error structure implied by $P(\hat{\kappa}_{i,t} \mid \kappa_{i,t})$.

$$L = \sum_{k \in \{D, R\}} P(\kappa_{i,t} = k) \cdot P(\hat{\kappa}_{i,t} \mid \kappa_{i,t} = k) \cdot P(\tilde{\psi}_{i,t} \mid \kappa_{i,t}, s_{i,t}) \quad (12)$$

For the mislabeling parameters, we impose Beta(1, 4) priors on the false-positive and false-negative rates (e_0, e_1) , reflecting the prior belief that misclassification is relatively rare, and a symmetric Beta(2, 2) prior on the population share of Democrats p . All belief-level parameters—intercepts $(\alpha^{up}, \alpha^{down})$, slopes $(\lambda^{up}, \lambda^{down})$, and the partisan-bias term (ω) —are assigned weakly informative $\mathcal{N}(0, 2)$ priors, while the residual standard deviation σ follows a HalfNormal(2)

prior. Posterior inference is conducted using the No-U-Turn Sampler (NUTS) with four parallel chains, each run for 1,500 warmup iterations followed by 1,500 post-warmup draws, and a target acceptance rate of 0.95.

Lastly, we estimate the model for each election cycle separately so that we can assess whether the parameters implied by the survey data actually vary over time. For six elections with no partisan switch, relying on a 4-year window around each election cannot identify belief changes attributable to changes in $x_{i,t}$. For these non-switch cycles, we extended our sample backward by one or two cycles to include a political turnover.

4.4 Results

For each of the 12 U.S. presidential election cycles from 1980 to 2024, we estimate all model parameters for three pairs of micro and macro expectations elicited in MSC.²¹ The variable pairs include the following: MSC sentiment about personal finance *micro_sent* versus macroeconomy *macro_sent*; personal finance expectations (*perp*) versus overall business conditions (*beexp*); real income expectations (*rinc*) versus business conditions (*bus12*).

The estimated parameters are reported in Figure 7. Each diagram corresponds to one of the three: $\lambda^{up} - \lambda^{down}$, $\alpha^{up} - \alpha^{down}$, and ω , and contains the estimates for all 12 election cycles and for 3 expectation pairs. Their highest-density interval (HDI) between 10% and 90% from the posterior distribution is plotted. Unconverged estimations are separately flagged as gray bars.

With the exception of the 1996 election cycle, three patterns consistently emerge from our estimates. First, the estimates of $\alpha^{up} - \alpha^{down}$ indicate that, across expectation domains and election cycles, the supporters of the incumbent president’s party are more optimistic about the macroeconomy than those of the opposition party. This is implied by consistently positive differences between α^{up} and α^{down} across all expectation pairs, with higher expectation values corresponding to a more positive outlook.

Second, it seems that, across measures and years, a predicted Democrat holds less positive views about the macroeconomy than a predicted Republican, which is implied by the negative estimate of ω . Overall, given the measured heterogeneity in microeconomic expectations, all three components, as defined above, contribute to partisanship in expectations during our estimation sample. The estimates suggest the overall importance of the state-dependent nature of the partisanship as implied by the non-identical λ ’s and α ’s. Partisanship is not simply a time-invariant trait across individuals; instead, it reflects its interactions with national politics.

Third, across specifications, supporters of the incumbent and challenger also differ in their

²¹We focus on MSC due to its long historical span.

degree of belief extrapolation, i.e., λ^{up} generally differing from λ^{down} . Although the exact way belief extrapolation varies across domains of expectations is unclear, the estimates for the 2016 to 2024 election cycles imply that the mapping from microeconomic expectations to macroeconomic views is more intensified for incumbent parties. We can interpret these results as households supporting the opposition party holding onto their priors and reacting less to their new information in forming their views about the macroeconomy.

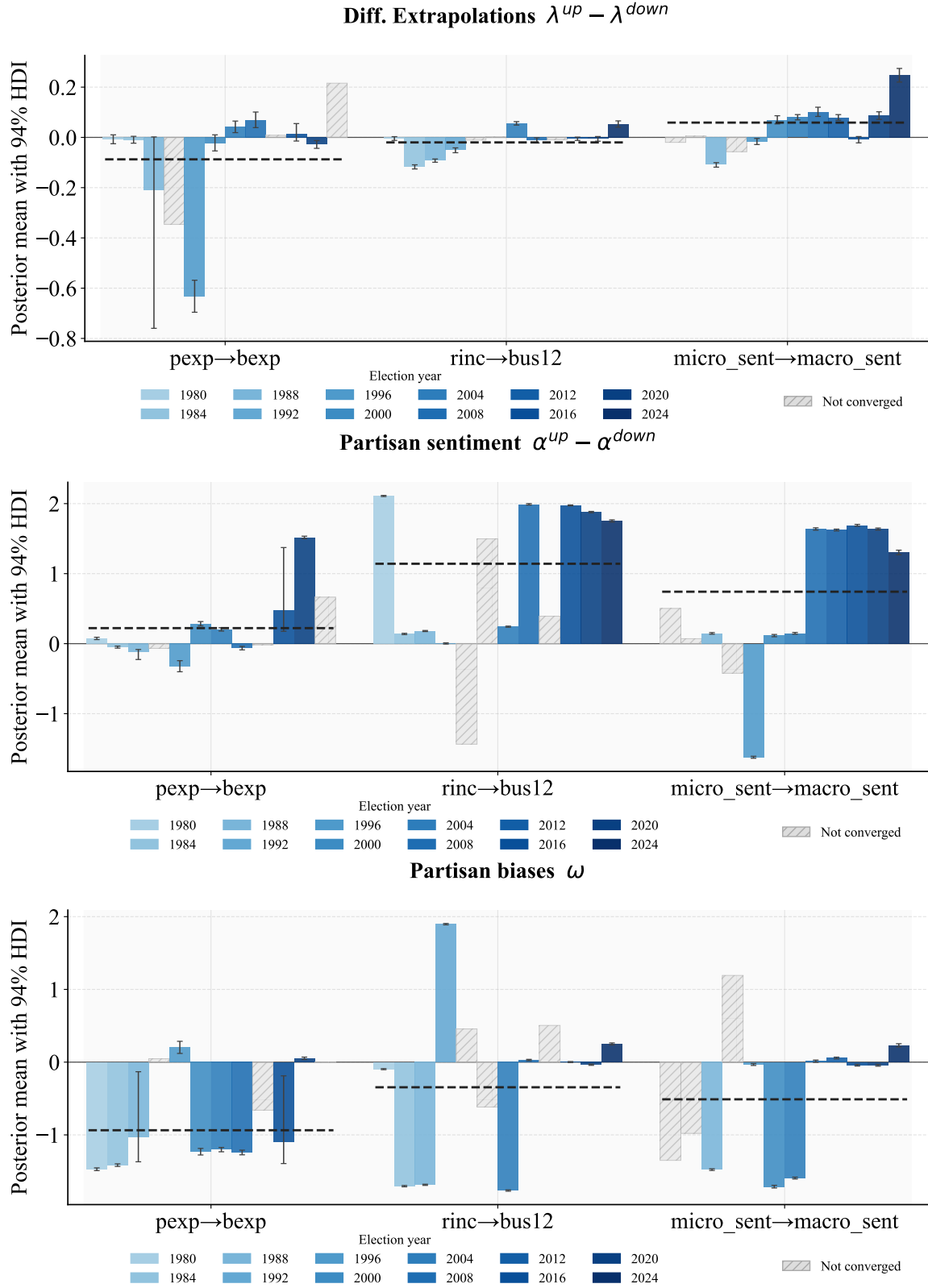
Across all election cycles, 1992 and 1996 stand out as caveats to the overall pattern. But there seem to be good reasons for this. For instance, Figure 3d shows that during these periods, real income expectations, which we treat as microeconomic expectations, actually experienced a sudden switch around midterm elections, 2 years before the presidential race. This sudden positive shift in Democrats’ personal real income expectations would predict macroeconomic expectations that are too optimistic. This pushes the implied α estimates in the opposite direction compared to other times.

Time-varying importance of partisanship: Once the model is estimated, we evaluate the time-varying importance of each mechanism in generating cross-sectional differences in macroeconomic expectations. According to our model, polarized views about the macroeconomy stem from both dispersed microeconomic conditions ($s_{i,t}$) of different households, and the “colored” glasses which induce distorted views of the same micro reality.²² We assess the relative contribution of these two channels as well as the overall importance of the model mechanisms in explaining the macroeconomic expectations’ heterogeneity.

Our decomposition is implemented through a sequential shutdown procedure applied to 500 posterior draws of all parameters. For each draw, the full parameter vector is used to compute the within-month cross-sectional variance of model-predicted macro belief $\hat{\psi}_{i,t}$. Channel contributions are identified by setting parameters to neutral values one at a time (e.g., $\omega = 0$, $\alpha^{up} = \alpha^{down}$, $\lambda^{up} = \lambda^{down}$) and measuring the resulting change in variance relative to the previous step. Since the model explicitly yields estimates of labeling error rates and population-type probabilities, we use these estimates to correct the imputed label based on Bayes’ rule. Our variance decomposition accounts for the uncertainty of the measures. Figure 8 presents the variance decomposition by election cycle for each pair of expectations. The bar chart displays the relative contribution of each channel to the model-implied variance of macroeconomic expectations (excluding the residual component). The line chart below reports the absolute, time-varying share attributable to partisan mechanisms. Consequently, one minus the plot-

²²As one example of how the increasing economic inequality leads to polarized politics, see Autor et al. (2020). The paper documents the political polarization as partly a consequence of the increasing economic inequality experienced by impacted workers. Although our focus is primarily on short-term personal economic conditions, it is sensible to assume that more persistent/permanent income inequality may enter macroeconomic expectations in ways similar to our $s_{i,t}$.

Figure 7: Parameter estimates by election cycle



Notes: These figures plot, for 12 election cycles, the belief parameter estimates based on their posterior modes and their 10% and 90% highest density intervals. The average estimates across all cycles is plotted as a dashed line. Gray bars flag non-converged estimates.

ted share corresponds to the period-specific residual variance that remains unexplained by the model.

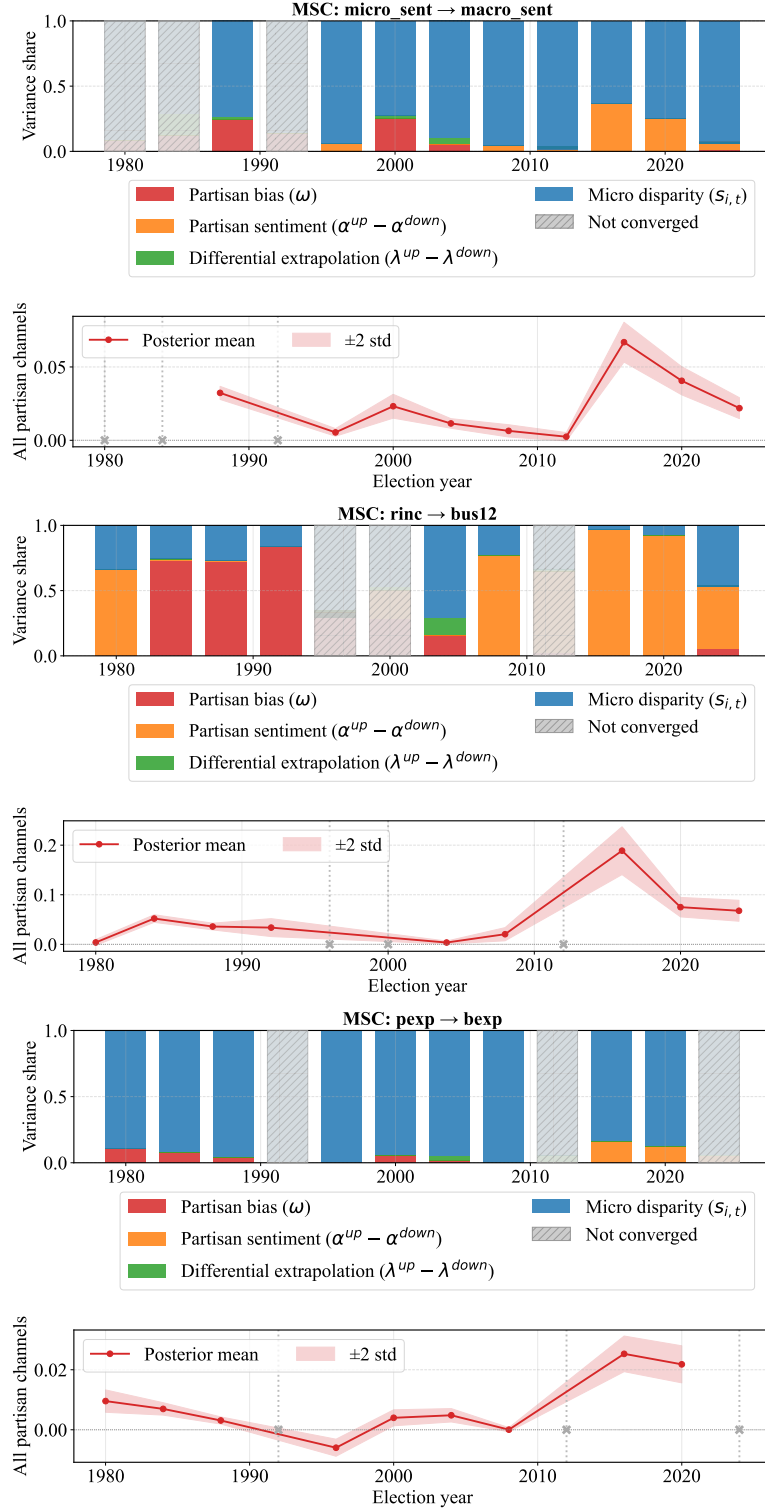
According to the decomposition, the relevance of different partisan channels varies over time. During the 1980 election and again in the post2016 periodphases marked by substantial partisan switching in proximity to electionsthe partisan sentiment channel emerges as the predominant explanatory factor. By contrast, in the elections held between 1984 and 1992, partisan bias constituted the primary source of partisan influence. The contribution of partisan extrapolation is relatively limited, although it remains empirically relevant. When all partisan channels are deactivated, the residual variation is accounted for by dispersion in households micro-level expectations, $s_{i,t}$, which remains consistently large across elections and variables.

The results also suggest that the partisan influences on macroeconomic expectations are not purely a recent phenomenon, although they have intensified markedly in the three election cycles since 2016. This pattern aligns with recent evidence emphasizing polarization during the 2016 and 2020 elections. 2016 did stand out as the most polarized election, with 20% of the variance share explained by the model. However, model estimates from earlier periods, particularly since 2008, also show meaningful contributions from partisan factors across several expectations. An earlier episode of relatively elevated partisan influence occurred in the 1980s, though at levels below those observed in the past decade. By contrast, during the three intervening decades, partisanship plays only a limited role in explaining heterogeneity in macroeconomic expectations.

Finally, it is important to note that the micro disparity captured by $s_{i,t}$ does not fully reflect the importance of personal circumstances in shaping macroeconomic beliefs. A large share of the variation in expectations remains unexplained by the model, likely reflecting unobserved heterogeneity and more persistent differences in households economic conditions and outlooks. This reinforces the paper’s central theme: despite the growing politicization of macroeconomic expectations, most of the observed dispersion cannot be attributed to partisan channels alone.

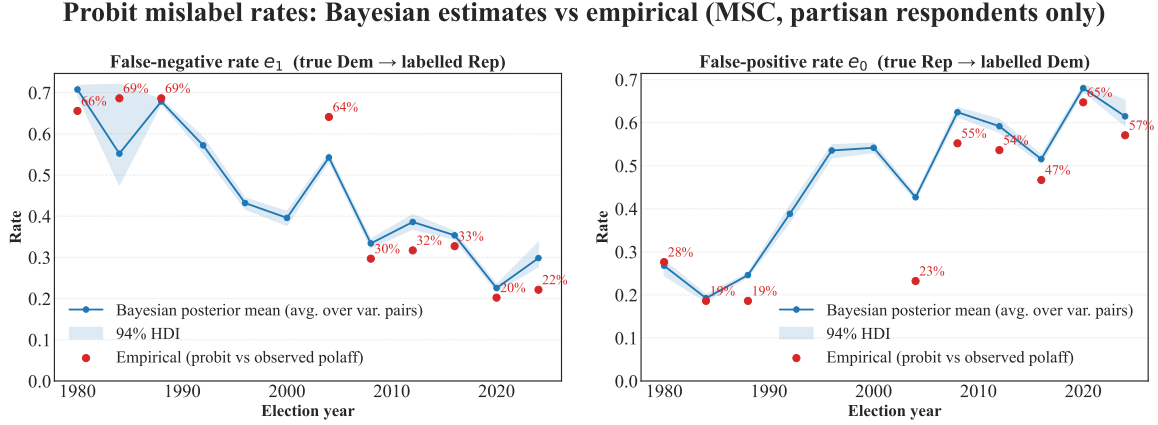
Estimated mislabel rates versus actual mislabel rates. With estimates of the mislabeling rates $\hat{e}_0$ and $\hat{e}_1$, we can evaluate the model’s out-of-sample performance in an indirect manner. That is to compare the model-implied mislabel rates with the actual mislabeling rates. The latter is calculated for imputations $\hat{\kappa}_{i,t}$ relative to the actual partisan affiliations, which are mostly available since 2016. This is an out-of-sample validation of the belief model, in that the model-based error rates are inferred indirectly from the belief patterns, without relying on the MSC’s actual labels. Remember that our imputed types $\hat{\kappa}_{i,t}$ are based on a different data set, ANES, and such labels are not all part of in-sample training. Therefore, the observed mislabeling rates are truly external benchmarks for assessing such model-based inference. Figure 9 plots such a comparison. The estimated error rates for each election cycle are averaged across

Figure 8: Decompositions of Dispersion in Macroeconomic Expectations: Heterogeneity: Partisanship and Micro Disparity



Notes: These figures plot the variance contribution of each partisan channel to the within-election-cycle variations of the macroeconomic expectations. The variance contribution is obtained by sequentially shutting down each channel in 500 draws from the posterior distribution. Non-converged estimations are flagged.

Figure 9: Mislabel rates: model estimation versus actual



variable pairs. Surprisingly, the two sets of error rates align quite closely, especially for the recent three election cycles.

Note, meanwhile, that both error rates are not always low. The probability of mislabeling a true Democrat in the early sample and mislabeling a Republican as a Democrat in the later sample could be as high as 70%. This does not invalidate our model’s specifications per se, but instead suggests that overall, the imputation is a rather noisy measure of one’s true political affiliation. This also highlights the importance of explicitly modeling mislabeling in any downstream estimation with the generated partisan affiliations.

5 Inspecting the Role of Policy Expectations Driving Partisan Macroeconomic Expectations

Our evidence confirms that households hold partisan views of the economy. However, we remain agnostic about the underlying causes of the so-called “colored glasses” until this point. In addition to previous explanations by the literature, such as the polarization of news media along partisan lines²³, we inspect the role of households’ expectations of public policies.

Political alignment undoubtedly stems from a wide range of non-economic considerations. Still, as economists, we test the hypothesis that households with different expectations on the positive(negative) macroeconomic effects of policies enacted by the winning presidential candidate can explain sudden shifts in optimistic(pessimistic) shifts of partisan expectations shortly after elections. We find mixed evidence regarding such a hypothesis in Section 5.1. Households only positively associate some policies of their preferred winning party to a better macroeconomy but negatively associate others, which suggests that partisanship cheerleading

²³See evidence from [Chahrour et al. \(2024\)](#); [Cummings et al. \(2024\)](#); [Harris and Sojourner \(2024\)](#).

cannot be the only driver of policy views. Meanwhile, in Section 5.2, we report that self-interest remains prominent in driving households’ views of a policy’s impacts on the macroeconomy. Households tend to believe that policies which are *personally* beneficial are also beneficial to the *macroeconomy*. Our findings of the importance of both self-interest and partisanship echo the framework of Stantcheva (2020, 2024). We now turn to detailed results.

To explore the role of households’ beliefs on public policies on partially, we use data from the Public Policy Survey (PPS) sub-module of the SCE, which is conducted every 4 months. As an example, we look at partisan expectations concerning public policies during the 2016 and 2020 elections. First, we describe the SCE questions on household expectations of public policies. The PPS asks households on two types of questions public policy expectations: first on which direction a respondent thinks that a given policy will change, and second on whether the respondent thinks this presumed policy change will have a positive/negative effect on their own household.²⁴ We use the PPS submodule’s microdata for 6 types of public policies: welfare benefits, payroll tax increase, unemployment benefits, capital gains tax, income tax, and income tax for the highest income bracket.

We then use these variables to construct proxy variables, for each type of public policy, on the household’s expected “Direction” of policy change and the expected “Effect” of that policy on their individual household. We construct our dataset from the PPS variables as such. We adjust each policy *direction* variable into a $[-1, 1]$ scale where $-1(1)$ indicates that the respondent expects the policy to be reduced/decreased(expanded/increased). We also adjust each policy effect variable, also to a $[-1, 1]$ scale, where -1 indicates that a respondent thinks that the direction of the policy change will have a very negative effect on their household, and 1 for very positive effect. The PPS submodule specifically asks for effects to the responding household and not the wider economy, which allows us to directly observe the policy’s effects on households’ micro expectations.

5.1 Partisan Cheerleading and Policy Expectations

In this subsection, we aim to see how partisan households adjust their micro and macro beliefs conditional on their preferred political candidate winning the presidency. Here, for each policy j , we run regressions with the Direction variable $\text{PolicyDirection}_{jit}$ and its interaction terms with the Dem_i , Win_{it} , and PostElect_t variables. Specifically, we look households’ responses in the 2016, 2020, and 2024 election cycles, using the full available SCE microdata. We run the following policy direction belief regressions, separately for each policy j , with control variables and election cycle dummy variables:

²⁴For the latter type of variables, the PPS questions specifically ask for the policy change’s effect on the *individual* household, not the general public.

$$\begin{aligned}
\text{Belief}_{it} = & \alpha_0 + \alpha_1 \text{Dem}_i + \alpha_2 \text{Win}_{it} + \alpha_3 \text{PostElect}_t + \alpha_4 \text{Win}_{it} \times \text{PostElect}_t \\
& + \alpha_5 \text{PolicyDirection}_{jit} + \alpha_6 \text{PolicyDirection}_{jit} \times \text{Dem}_i \\
& + \alpha_7 \text{PolicyDirection}_{jit} \times \text{Win}_{it} + \alpha_8 \text{PolicyDirection}_{jit} \times \text{PostElect}_t \\
& + \alpha_9 \text{PolicyDirection}_{jit} \times \text{Win}_{it} \times \text{PostElect}_t \\
& + \text{Controls}_i + \varepsilon_{it}
\end{aligned} \tag{13}$$

In Table A.21 of the Online Appendix, we show the signs and coefficients from these regressions for the same 6 policies mentioned above. In that table, the coefficients in the “Direction” column are our α_4 estimates, namely the correlation between the direction of the policy change and the change in macro/micro beliefs. The “ $\times \text{Dem}_i$ ” column corresponds to α_6 , namely the additional effect of Democratic partisanship on macro/micro beliefs. The “ $\times \text{Win}_{it}$ ” column corresponds to α_7 , the bias between households who support the winning candidate and households who do not. The “ $\times \text{postElect}_t$ ” column corresponds to α_8 , the post-electoral shift in households’ economic beliefs. Finally, “ $\times \text{Win}_{it} \times \text{postElect}_t$ ” column corresponds to α_9 , namely the post-electoral partisan effect on households’ beliefs after the election victory of the candidate that the household prefers.

One finding of interest in Table A.21 is on the $\times \text{Dem}_i \times \text{postElect}_t$ coefficients, i.e. the change in macro/micro beliefs for Democratic households after each election. The $\times \text{Dem}_i \times \text{postElect}_t$ column (α_8) is generally, if significant, negative. In the case for beliefs on the probability of a higher national unemployment rate, we observe all 6 out of 6 of these coefficients as statistically insignificant or zero. However, we also see that this coefficient has more null results for the perceived job separation probability (5 zero coefficients, 1 negative coefficient).

This implies that, for example after Joe Biden’s election in November 2020, Democrats’ beliefs of the macroeconomy (the national unemployment rate) generally decreased by more than Republicans’ beliefs, but much less so for their own personal economic situation (the perceived probability of job separation). The evidence implies a partisan and post-electoral effect of households’ macro beliefs, but less so in micro beliefs; Democrats who saw Joe Biden win the 2020 election thought that the economy would improve (lower national unemployment) even if their own circumstances would not change. Thus, our results in Table A.21 point to our main finding again, that electoral turnover are more likely to affect households’ beliefs about the macroeconomy than those about their personal economic situation.

5.2 Self-interested Motives in Policy Views

In this subsection, we look into self-interested motives as a possible cause of the partisan shift in micro and macro expectations. We run regressions for each policy to a select number of macro/micro belief variables Belief_{it} , with control variables and an election cycle dummy variable,²⁵ separately for each policy j :

$$\text{Belief}_{it} = \alpha_0 + \alpha_1 \text{PolicyDirection}_{jit} + \alpha_2 \text{PolicyEffect}_{jit} + \text{Controls}_i + \varepsilon_{it}$$

where α_1 represents how much individuals' beliefs on expected direction of policy changes affects macro/micro beliefs, and α_2 represents the same for expected effects of policy changes.

In Table A.22 of the Online Appendix, we show the signs and significance levels of each policy belief coefficient in each regression involving a macro or micro expectation variable. For a visual summary, we include bar plots, with standard error bands, for 4 variables in Figure 10. We include results for expectations for the national unemployment rate, the individual job separation rate, as well as the probability of inflation increasing, and the expected increase in gas prices, all in the next 12 months. Each bar represents the estimated coefficient of the policy belief variable on a macro/micro belief variable.

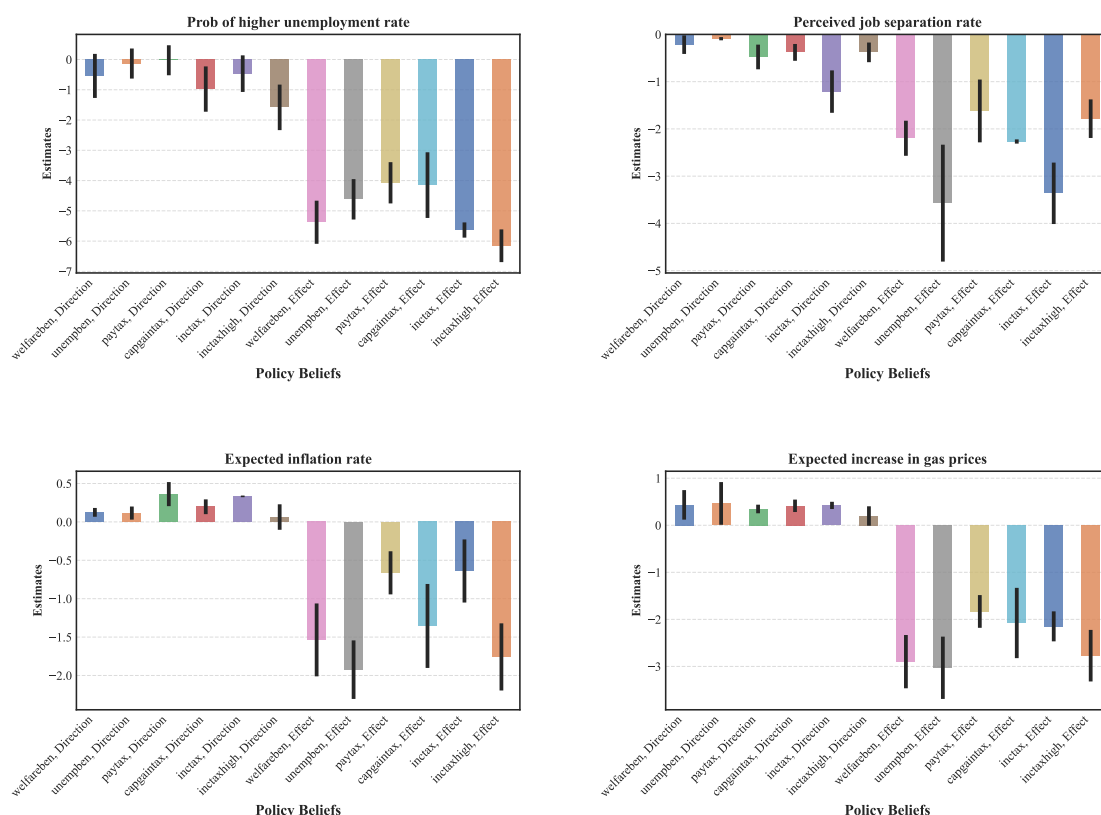
It is visually striking that the coefficients for the vast majority of the “Effect” variables, regardless of the policy (and of the direction of change of policy), are highly significant and exhibit the same sign. For example, for the national unemployment rate, all the Effect variables (the right half of the graph) are negative and statistically significant. We also see this for expectations of individuals' job separation rates and also for expectations of the inflation rate and of gas prices. In other words, individuals who expect policy changes that will have a positive effect on their households also tend to expect macroeconomic conditions to improve in the future (lower unemployment, lower inflation, etc.). Generally, we interpret these results as preliminary evidence that individuals relate their expectations on the effects of policies to expectations of macro and micro variables. We take these results as supporting evidence to our claim that households' expectations of their personal circumstances are positively correlated with their expectations of the macroeconomy.

6 Conclusion

A rapidly growing literature highlights the role of partisanship in shaping households macroeconomic expectations. Our findings show that while such partisan patterns are evident in macro

²⁵Given that the SCE currently only has data for the 2016, 2020, and 2024 election cycles, we include dummy variables for each election cycle.

Figure 10: Estimates of Coefficients from Policy Belief Regressions



Notes: Bar plot of estimated coefficients from policy belief regressions, for select macro/micro belief variables. Each thick colored bar is the estimated coefficient in the corresponding regression for the policy, with ± 1 standard deviation around the mean in thin black bars. The left half of each bar plot shows the estimated coefficients for the policy belief *direction* variables, and the right half for the policy belief *effect* variables.

beliefs, they are much less pronounced in households expectations about their own economic futures—likely because individuals are better informed about their personal circumstances. We document this distinction across a wide array of beliefs using data from both the SCE and MSC surveys. Leveraging imputed political preferences spanning 12 U.S. election cycles (1980-2024), we further show that partisan gaps in macroeconomic views vary over time, with particularly sharp divides in the 1980s and after 2016.

To explain these patterns, we propose a model of belief formation—what we term “seeing the economy through colored glasses.” In this framework, individuals directly observe their personal economic realities and project them onto the broader economy, with partisanship shaping how this extrapolation occurs. Empirical estimates across multiple pairs of micro and macro expectations confirm that partisanship affects not only the level of macroeconomic beliefs, but also the way individuals map personal experiences onto national economic assessments.

Our central conclusion is that, despite partisan biases, personal experience remains the foun-

dation of macroeconomic beliefs. In line with this, we show that individuals views on macroeconomic policy are also shaped by self-interested extrapolation: rather than evaluating policies based on their aggregate effects, people interpret them through the lens of their own potential gains or losses. Consistent with this mechanism, consumption responses and in turn macroeconomic dynamics appear to be driven primarily by these microeconomic beliefs rather than by partisan shifts in macroeconomic expectations.

References

- Abril-Pla, Oriol, Virgile Andreani, Colin Carroll, Larry Dong, Christopher J. Fonnesebeck, Maxim Kochurov, Ravin Kumar, Junpeng Lao, Christian C. Luhmann, Osvaldo A. Martin, Michael Osthege, Ricardo Vieira, Thomas Wiecki, and Robert Zinkov**, “PyMC: A Modern and Comprehensive Probabilistic Programming Framework in Python,” *PeerJ Computer Science*, 2023, 9 (e1516).
- Addoum, Jawad M and Alok Kumar**, “Political sentiment and predictable returns,” *The Review of Financial Studies*, 2016, 29 (12), 3471–3518.
- American National Election Studies**, “ANES 2020 Time Series Study Full Release [dataset and documentation],” 2021.
- Andre, Peter, Carlo Pizzinelli, Christopher Roth, and Johannes Wohlfart**, “Subjective models of the macroeconomy: Evidence from experts and representative samples,” *The Review of Economic Studies*, 2022, 89 (6), 2958–2991.
- Autor, David, David Dorn, Gordon Hanson, and Kaveh Majlesi**, “Importing political polarization? The electoral consequences of rising trade exposure,” *American Economic Review*, 2020, 110 (10), 3139–3183.
- Bachmann, Oliver, Klaus Gründler, Niklas Potrafke, and Ruben Seiberlich**, “Partisan bias in inflation expectations,” *Public Choice*, 2021, 186, 513–536.
- Bartels, L. M.**, “Beyond the Running Tally: Partisan Bias in Political Perceptions,” *Political Behavior*, 2002, 24, 117–150.
- Binder, Carola**, “Political party affiliation and inflation expectations,” *Working Paper*, 2023.
- Binder, Carola Conces, Rupal Kamdar, and Jane M Ryngaert**, “Partisan expectations and COVID-era inflation,” *Journal of Monetary Economics*, 2024, p. 103649.

- Bolhuis, Marijn A, Judd NL Cramer, Karl Oskar Schulz, and Lawrence H Summers**, “The Cost of Money is Part of the Cost of Living: New Evidence on the Consumer Sentiment Anomaly,” Technical Report, National Bureau of Economic Research 2024.
- Bordalo, Pedro, Nicola Gennaioli, Florencio Lopez de Silanes, Simon G Schröder, Andrei Shleifer, and Maarten van Rooij**, “The Psychology of Macroeconomic Expectations,” Technical Report, National Bureau of Economic Research 2026.
- Born, Benjamin, Zeno Enders, Manuel Menkhoff, Gernot J Müller, and Knut Niemann**, “Firm expectations and news: micro v macro,” *Working Paper*, 2022.
- Brown, Jacob R, Enrico Cantoni, Ryan Enos, Vincent Pons, and Emilie Sartre**, “Sources and Extent of Rising Partisan Segregation in the US—Evidence from 143 Million Voters,” Technical Report, National Bureau of Economic Research 2025.
- Brunnermeier, Markus K and Jonathan A Parker**, “Optimal expectations,” *American Economic Review*, 2005, *95* (4), 1092–1118.
- Bullock, John G., Alan S. Gerber, Seth J. Hill, and Gregory A. Huber**, “Partisan Bias in Factual Beliefs about Politics,” *Quarterly Journal of Political Science*, 2015, *10*, 519–578.
- Bullock, John G and Gabriel Lenz**, “Partisan bias in surveys,” *Annual Review of Political Science*, 2019, *22* (1), 325–342.
- Burke, Mary A and Ali Ozdagli**, “Household inflation expectations and consumer spending: evidence from panel data,” *The Review of Economics and Statistics*, 2023, *105* (4), 948–961.
- Burn-Murdoch, John**, “How our sense of economic reality is being distorted,” *Financial Times* May 2024.
- Bursztyn, Leonardo, Aakaash Rao, Christopher Roth, and David Yanagizawa-Drott**, “Opinions as facts,” *The Review of Economic Studies*, 2023, *90* (4), 1832–1864.
- Campbell, Angus, Philip E. Converse, Warren E. Miller, and Donald E. Stokes**, *The American Voter*, John Wiley & Sons., 1960.
- Casselman, Ben**, “Republicans See a Better Economic Outlook. Now Its Democrats Who Dont.,” *The New York Times* November 2024. Accessed: 2025-01-31.
- Cassidy, Will and Blair Vorsatz**, “Partisanship and portfolio choice: Evidence from mutual funds,” *Available at SSRN 3977887*, 2021.
- Chahrour, Ryan, Adam Hale Shapiro, and Daniel J Wilson**, “News Selection and Household Inflation Expectations,” Technical Report 2024.

Chinoy, Sahil, Nathan Nunn, Sandra Sequeira, and Stefanie Stantcheva, “Zero-sum thinking and the roots of US political divides,” Technical Report, National Bureau of Economic Research 2023.

Choi, Sangyup, Sang-Hyun Kim, Myunghwan Andrew Lee, Siye Bae, and Myungkyu Shim, “Partisan Bias in Inflation Beliefs: New Evidence from Korea,” Technical Report 2022.

Conover, Pamela Johnston, Stanley Feldman, and Kathleen Knight, “The personal and political underpinnings of economic forecasts,” *American Journal of Political Science*, 1987, pp. 559–583.

Cookson, J Anthony, Joseph E Engelberg, and William Mullins, “Echo chambers,” *The Review of Financial Studies*, 2023, 36 (2), 450–500.

Crump, Richard K, Stefano Eusepi, Andrea Tambalotti, and Giorgio Topa, “Subjective intertemporal substitution,” *Journal of Monetary Economics*, 2022, 126, 118–133.

Cummings, Ryan and Neale Mahoney, “Asymmetric amplification and the consumer sentiment gap,” Nov 2023. Briefing Book blog post.

—, **Giacomo Fraccaroli, and Neale Mahoney**, “Bad News Bias in Gasoline Price Coverage,” May 2024. Briefing Book blog post.

Dagostino, Ramona, Janet Gao, and Pengfei Ma, “Partisanship in loan pricing,” *Journal of Financial Economics*, 2023, 150 (3), 103717.

Dräger, Lena and Giang Nghiem, “Are consumers’ spending decisions in line with a euler equation?,” *Review of Economics and Statistics*, 2021, 103 (3), 580–596.

Drobot, Sergii, Daniela Puzzello, Ryan Rholes, and Alena Wabitsch, “Incentivizing inflation expectations,” *Available at SSRN 5226305*, 2024.

Du, William, Adrian Monninger, Xincheng Qiu, and Tao Wang, “Perceived Unemployment Risks over Business Cycles,” Technical Report, Bank of Canada 2025.

Duca-Radu, Ioana, Geoff Kenny, and Andreas Reuter, “Inflation expectations, consumption and the lower bound: Micro evidence from a large multi-country survey,” *Journal of Monetary Economics*, 2021, 118, 120–134.

Economist, The, “The pandemic has broken a closely followed survey of sentiment,” *The Economist* September 2023.

—, “American consumers are finally cheering up,” *The Economist* February 2024.

- , “Why are Americans so gloomy about their great economy?,” *The Economist* January 2024.
- Engelberg, Joseph, Runjing Lu, William Mullins, and Richard R Townsend**, “Political sentiment and innovation: Evidence from patenters,” Technical Report, National Bureau of Economic Research 2023.
- Evans, Mary Claire**, “U.S. Economic Confidence Ticks Down as Partisans’ Views Shift,” *Gallup News* January 2025. Accessed: 2025-01-31.
- Ferrario, Beatrice and Stefanie Stantcheva**, “Eliciting peoples first-order concerns: Text analysis of open-ended survey questions,” in “AEA Papers and Proceedings,” Vol. 112 American Economic Association 2014 Broadway, Suite 305, Nashville, TN 37203 2022, pp. 163–169.
- Gaines, Brian J, James H Kuklinski, Paul J Quirk, Buddy Peyton, and Jay Verkuilen**, “Same facts, different interpretations: Partisan motivation and opinion on Iraq,” *The Journal of Politics*, 2007, 69 (4), 957–974.
- Gillitzer, Christian, Nalini Prasad, and Tim Robinson**, “Political attitudes and inflation expectations: Evidence and Implications,” *Journal of Money, Credit and Banking*, 2021, 53 (4), 605–634.
- Guillochon, Justine**, “News-driven household macroeconomic expectations: regional vs. national telecast information,” *Available at SSRN 4071206*, 2022.
- Harris, Ben and Aaron Sojourner**, “Why are Americans so displeased with the economy?,” 2024.
- Ichiiue, Hibiki and Shusaku Nishiguchi**, “Inflation expectations and consumer spending at the zero bound: Micro evidence,” *Economic Inquiry*, 2015, 53 (2), 1086–1107.
- Jeong, Jaemin, Soojin Jo, and Myungkyu Shim**, “When News Isn’t Just News: Partisan Affiliation and Inflation Expectations,” *Available at SSRN 5239703*, 2025.
- Kamdar, Rupal and Walker Ray**, “Polarized Expectations, Polarized Consumption,” 2022.
- Kempf, Elisabeth and Margarita Tsoutsoura**, “Partisan professionals: Evidence from credit rating analysts,” *The Journal of Finance*, 2021, 76 (6), 2805–2856.
- Kuang, Pei, Michael Weber, and Shihan Xie**, “Perceived Political Bias of the Federal Reserve,” Technical Report, National Bureau of Economic Research 2024.
- McCartney, W Ben, John Orellana-Li, and Calvin Zhang**, “Political polarization affects households’ financial decisions: evidence from home sales,” *The Journal of Finance*, 2024, 79 (2), 795–841.

- Meeuwis, Maarten, Jonathan A Parker, Antoinette Schoar, and Duncan Simester**, “Belief disagreement and portfolio choice,” *The Journal of Finance*, 2022, 77 (6), 3191–3247.
- Mian, Atif, Amir Sufi, and Nasim Khoshkhoh**, “Partisan bias, economic expectations, and household spending,” *Review of Economics and Statistics*, 2021, pp. 1–46.
- of Consumers University of Michigan, Survey**, “Survey of Consumers - Notes: The Partisan Economy,” 2022.
- , “Survey of Consumers - Notes: Partisan Perceptions and Expectations,” 2024.
- Prior, Markus, Gaurav Sood, Kabir Khanna et al.**, “You cannot be serious: The impact of accuracy incentives on partisan bias in reports of economic perceptions,” *Quarterly Journal of Political Science*, 2015, 10 (4), 489–518.
- Rice, Anthony B**, “Executive partisanship and corporate investment,” *Journal of Financial and Quantitative Analysis*, 2020, pp. 1–30.
- Stantcheva, Stefanie**, “Understanding economic policies: What do people know and learn,” *Unpublished Manuscript, Harvard University*, 2020, pp. 2309–2369.
- , “Why do we dislike inflation?,” Technical Report, National Bureau of Economic Research 2024.
- Taubinsky, Dmitry, Luigi Butera, Matteo Saccarola, and Chen Lian**, “Beliefs about the economy are excessively sensitive to household-level shocks: Evidence from linked survey and administrative data,” Technical Report, National Bureau of Economic Research 2024.
- Vilbig, Karyn and Paula England**, “Elites moved toward democrats more than nonelites moved away: Income, education, and occupational class in US presidential elections, 1980–2020,” *Proceedings of the National Academy of Sciences*, 2026, 123 (10), e2526263123.
- Wang, Tao**, “Perceived versus calibrated income risks in heterogeneous-agent consumption models,” Technical Report, Bank of Canada 2023.

A Additional Empirical Results

A.1 Probit model coefficient estimates for each survey and year

1980 (MSC)	Democrat		Republican	
Age	-0.0084	(0.0111)	0.0062	(0.0114)
Age squared	0.0001	(0.0001)	0	(0.0001)
Female	0.2201***	(0.0813)	-0.249***	(0.0795)
HS or less	0.3029***	(0.1108)	-0.343***	(0.1121)
Some College	0.084	(0.1113)	-0.1168	(0.1096)
BA Degree	-0.2776**	(0.136)	0.1826	(0.1264)
Post BA	0.1791	(0.1736)	-0.1503	(0.1717)
Married	-0.0375	(0.0911)	0.1951**	(0.0897)
3+ children	0.3223**	(0.1305)	-0.243*	(0.1334)
Income Mid	-0.1775	(0.1115)	0.2167*	(0.113)
Income High	-0.2751**	(0.1239)	0.3221***	(0.1234)
Constant	-0.264	(0.2515)	-0.5301**	(0.259)
Nobs	1,109		1,109	
R2	0.042		0.046	

Table A.1: Coefficients of demographic variables from Probit regression of Democratic/Republican party affiliation using demographic variables common to American National Election Survey (ANES) and Michigan Survey of Consumers (MSC) cross-sectional data. 1980 Presidential election, with a sample period from Jan 1979 to Dec 1982.

A.2 Alternative imputation methods

Tables A.17 and A.18 report the imputation accuracies based on two alternative machine learning models. The model is trained and cross-validated with the pooled ANES data. Since the election year is included as a predictor and both models capture non-linear interactions between features, it naturally captures time-varying voter profiles along the considered demographic variables. Despite its flexibility, both models yield prediction accuracies comparable to our baseline.

A.3 Additional regression results

1984 (MSC)	Democrat		Republican	
Age	0.0208**	(0.0093)	-0.0072	(0.0093)
Age squared	-0.0002**	(0.0001)	0.0001	(0.0001)
Female	0.1464**	(0.0601)	-0.107*	(0.0598)
HS or less	0.262***	(0.0845)	-0.3178***	(0.085)
Some College	0.0607	(0.0803)	0.0224	(0.0794)
BA Degree	0.116	(0.1003)	-0.0155	(0.0997)
Post BA	0.3794***	(0.1349)	-0.3762***	(0.1351)
Married	-0.149**	(0.0652)	0.2114***	(0.065)
3+ children	-0.1683	(0.1081)	0.121	(0.1074)
Income Mid	-0.2689***	(0.0776)	0.353***	(0.0778)
Income High	-0.4695***	(0.0906)	0.5923***	(0.0904)
Constant	-0.4243**	(0.2101)	-0.1957	(0.2093)
Nobs	1,940		1,940	
R2	0.031		0.053	

Table A.2: Coefficients of demographic variables from Probit regression of Democratic/Republican party affiliation using demographic variables common to American National Election Survey (ANES) and Michigan Survey of Consumers (MSC) cross-sectional data. 1984 Presidential election, with sample period from Jan 1983 to Dec 1986.

1988 (MSC)	Democrat		Republican	
Age	0.0405***	(0.0111)	0.0243**	(0.0114)
Age squared	-0.0003***	(0.0001)	-0.0001	(0.0001)
Female	0.0931	(0.0669)	-0.0739	(0.067)
HS or less	-0.0125	(0.1249)	-0.3134**	(0.1351)
Some College	0.1124	(0.09)	0.3515***	(0.0892)
BA Degree	0.1298	(0.098)	0.5589***	(0.0946)
Post BA	0.5563***	(0.1353)	0.1461	(0.1379)
Married	0.029	(0.0725)	0.1336*	(0.0738)
3+ children	-0.1062	(0.118)	0.1245	(0.1196)
Income Mid	0.1172	(0.0833)	0.3107***	(0.0879)
Income High	-0.1377	(0.1051)	0.6925***	(0.105)
Constant	-1.6311***	(0.2608)	-1.8709***	(0.2677)
Nobs	1,803		1,803	
R2	0.028		0.104	

Table A.3: Coefficients of demographic variables from Probit regression of Democratic/Republican party affiliation using demographic variables common to American National Election Survey (ANES) and Michigan Survey of Consumers (MSC) cross-sectional data. 1988 Presidential election, with sample period from Jan 1987 to Dec 1990.

1992 (MSC)	Democrat		Republican	
Age	-0.0091	(0.0102)	0.0178*	(0.0101)
Age squared	0.0001	(0.0001)	-0.0001	(0.0001)
Female	-0.0514	(0.0572)	0.128**	(0.057)
HS or less	0.0228	(0.1156)	-0.025	(0.1151)
Some College	-0.0903	(0.0787)	0.0219	(0.0779)
BA Degree	0.1691**	(0.0836)	0.0219	(0.0843)
Post BA	-0.02	(0.1117)	0.2858***	(0.1099)
Married	0.1255*	(0.066)	-0.1417**	(0.0653)
3+ children	0.1844**	(0.0911)	-0.2698***	(0.0945)
Own Residence	0.01	(0.0645)	0.0442	(0.0643)
Income Mid	-0.0687	(0.0752)	-0.0569	(0.0746)
Income High	-0.0454	(0.0928)	-0.1636*	(0.0926)
Constant	-0.1839	(0.2393)	-0.7018***	(0.2381)
Obs.	2,112		2,112	
R2	0.008		0.018	

Table A.4: Coefficients of demographic variables from Probit regression of Democratic/Republican party affiliation using demographic variables common to American National Election Survey (ANES) and Michigan Survey of Consumers (MSC) cross-sectional data. 1992 Presidential election, with sample period from Jan 1991 to Dec 1994.

1996 (MSC)	Democrat		Republican	
Age	0.0275**	(0.0117)	-0.0143	(0.0125)
Age squared	-0.0003**	(0.0001)	0.0002**	(0.0001)
Female	0.2141***	(0.0671)	-0.1589**	(0.0702)
HS or less	0.1545	(0.1842)	-0.5855**	(0.2287)
Some College	-0.3023***	(0.0901)	0.2789***	(0.0939)
BA Degree	-0.2672***	(0.0938)	0.3786***	(0.0964)
Post BA	-0.049	(0.1225)	0.2011	(0.1256)
Married	-0.2077***	(0.0768)	0.1077	(0.0812)
3+ children	0.0042	(0.2369)	0.0176	(0.2545)
Own Residence	-0.1239	(0.0809)	0.0669	(0.086)
Income Mid	-0.2273**	(0.0896)	0.3987***	(0.0977)
Income High	-0.3703***	(0.1116)	0.6256***	(0.1189)
Constant	-0.0959	(0.2843)	-0.9602***	(0.3045)
Obs.	1,536		1,536	
R2	0.051		0.073	

Table A.5: Coefficients of demographic variables from Probit regression of Democratic/Republican party affiliation using demographic variables common to American National Election Survey (ANES) and Michigan Survey of Consumers (MSC) cross-sectional data. 1996 Presidential election, with sample period from Jan 1995 to Dec 1998.

2000 (MSC)	Democrat		Republican	
Age	0.0369**	(0.0155)	-0.045***	(0.0157)
Age squared	-0.0003**	(0.0001)	0.0004***	(0.0001)
Female	0.2944***	(0.083)	-0.1955**	(0.0833)
HS or less	0.0254	(0.2089)	-0.1677	(0.2191)
Some College	-0.1802*	(0.1052)	0.2671**	(0.1056)
BA Degree	-0.1957*	(0.1146)	0.1177	(0.1142)
Post BA	-0.0164	(0.144)	0.0882	(0.144)
Married	-0.1821**	(0.0917)	0.1469	(0.0931)
3+ children	0.0027	(0.1345)	0.0609	(0.1345)
Own Residence	-0.2421**	(0.0997)	0.3888***	(0.1026)
Income Mid	0.0039	(0.0973)	0.0192	(0.0981)
Income High	0.1479	(0.1159)	-0.0219	(0.1155)
Constant	-0.8795**	(0.4024)	0.513	(0.4046)
Obs.	1,059		1,059	
R2	0.039		0.039	

Table A.6: Coefficients of demographic variables from Probit regression of Democratic/Republican party affiliation using demographic variables common to American National Election Survey (ANES) and Michigan Survey of Consumers (MSC) cross-sectional data. 2000 Presidential election, with sample period from Jan 1999 to Dec 2002.

2004 (MSC)	Democrat		Republican	
Age	-0.0254	(0.016)	0.0349**	(0.0162)
Age squared	0.0002	(0.0002)	-0.0003**	(0.0002)
Female	0.0633	(0.0961)	0.0254	(0.0969)
HS or less	0.1554	(0.2126)	-0.233	(0.2186)
Some College	-0.03	(0.1235)	0.0632	(0.1241)
BA Degree	0.0568	(0.1284)	-0.056	(0.1286)
Post BA	0.5101***	(0.1545)	-0.5052***	(0.1556)
Married	-0.2157**	(0.1045)	0.343***	(0.1048)
3+ children	-0.3418	(0.2374)	0.3335	(0.2338)
Own Residence	0.0148	(0.1146)	0.0482	(0.1156)
Income Mid	-0.1435	(0.1092)	0.2355**	(0.1099)
Income High	-0.2044	(0.1632)	0.3403**	(0.1636)
Constant	0.6207*	(0.3626)	-1.3031***	(0.3703)
Obs.	780		780	
R2	0.029		0.052	

Table A.7: Coefficients of demographic variables from Probit regression of Democratic/Republican party affiliation using demographic variables common to American National Election Survey (ANES) and Michigan Survey of Consumers (MSC) cross-sectional data. 2004 Presidential election, with sample period from Jan 2003 to Dec 2006.

2008 (MSC)	Democrat		Republican	
Age	0.03	(0.022)	-0.0389*	(0.0226)
Age squared	-0.0004*	(0.0002)	0.0005**	(0.0002)
Female	0.0451	(0.1024)	0.0266	(0.1043)
HS or less	0.5821***	(0.2157)	-0.7769***	(0.236)
Some College	-0.0225	(0.1396)	0.0838	(0.1412)
BA Degree	0.1563	(0.1356)	-0.0809	(0.1369)
Post BA	0.2372	(0.1909)	-0.268	(0.1924)
Married	-0.5213*	(0.2965)	0.654**	(0.3258)
3+ children	-0.0999	(0.1574)	0.2913*	(0.1602)
Own Residence	-0.0174	(0.14)	0.1374	(0.145)
Income Mid	-0.2554	(0.1662)	0.3056*	(0.1735)
Income High	-0.7633***	(0.1823)	0.8769***	(0.1881)
Constant	0.5421	(0.5804)	-0.9408	(0.6006)
Obs.	667		667	
R2	0.065		0.091	

Table A.8: Coefficients of demographic variables from Probit regression of Democratic/Republican party affiliation using demographic variables common to American National Election Survey (ANES) and Michigan Survey of Consumers (MSC) cross-sectional data. 2008 Presidential election, with sample period from Jan 2007 to Dec 2010.

2012 (MSC)	Democrat		Republican	
Age	0.068***	(0.0219)	-0.0544**	(0.0224)
Age squared	-0.0007***	(0.0002)	0.0006***	(0.0002)
Female	-0.0805	(0.1028)	0.0527	(0.1057)
HS or less	0.5747***	(0.1824)	-0.8648***	(0.2088)
Some College	-0.1456	(0.146)	0.1983	(0.1483)
BA Degree	-0.1099	(0.143)	0.1913	(0.1443)
Post BA	0.2607	(0.1702)	-0.2443	(0.1738)
Married	-0.5071***	(0.1435)	0.6448***	(0.154)
3+ children	0.1856	(0.1997)	-0.1651	(0.2087)
Own Residence	-0.3092**	(0.1262)	0.2517*	(0.131)
Income Mid	-0.3078**	(0.1331)	0.2679*	(0.1396)
Income High	-0.3533**	(0.1575)	0.3191**	(0.1628)
Constant	-0.5314	(0.4937)	-0.1734	(0.5094)
Obs	661		661	
R2	0.075		0.101	

Table A.9: Coefficients of demographic variables from Probit regression of Democratic/Republican party affiliation using demographic variables common to American National Election Survey (ANES) and Michigan Survey of Consumers (MSC) cross-sectional data. 2012 Presidential election, with sample period from Jan 2011 to Dec 2014.

2016 (MSC)	Democrat		Republican	
Age	0.0088	(0.0087)	0.0093	(0.0088)
Age squared	-0.0001	(0.0001)	0	(0.0001)
Female	0.2474***	(0.0495)	-0.169***	(0.05)
HS or less	0.1977	(0.1267)	-0.165	(0.1287)
Some College	0.0748	(0.0701)	-0.02	(0.0701)
BA Degree	0.2468***	(0.0661)	-0.2236***	(0.0665)
Post BA	0.658***	(0.0759)	-0.5618***	(0.0775)
Married	-0.1309**	(0.0559)	0.1968***	(0.0563)
3+ children	-0.2399**	(0.1055)	0.1447	(0.1041)
Own Residence	-0.2902***	(0.0614)	0.2963***	(0.0622)
Income Mid	-0.0608	(0.0631)	0.1302**	(0.0641)
Income High	0.012	(0.0783)	0.0342	(0.0796)
Constant	-0.246	(0.2056)	-1.0045***	(0.2123)
Obs.	2,777		2,777	
R2	0.054		0.068	

Table A.10: Coefficients of demographic variables from Probit regression of Democratic/Republican party affiliation using demographic variables common to American National Election Survey (ANES) and Michigan Survey of Consumers (MSC) cross-sectional data. 2016 Presidential election, with sample period from Jan 2015 to Dec 2018.

2020 (MSC)	Democrat		Republican	
Age	-0.0166	(0.015)	0.0299*	(0.0153)
Age squared	0.0002	(0.0001)	-0.0002	(0.0001)
Female	0.0758	(0.0592)	-0.046	(0.0599)
HS or less	-0.2409	(0.1919)	-0.0334	(0.1786)
Some College	0.0611	(0.0915)	0.0378	(0.0892)
BA Degree	0.3677***	(0.0827)	-0.2619***	(0.0826)
Post BA	0.5688***	(0.0859)	-0.4963***	(0.0876)
Married	-0.1679*	(0.0908)	0.3146***	(0.0955)
3+ children	-0.2997***	(0.1045)	0.3212***	(0.1016)
Own Residence	-0.1393*	(0.0833)	0.1417*	(0.0855)
Income Mid	0.1022	(0.0911)	0.1232	(0.0906)
Income High	0.1761*	(0.0953)	-0.0204	(0.0955)
Constant	0.2334	(0.3611)	-1.6531***	(0.3748)
Obs	1,925		1,925	
R²	0.050		0.056	

Table A.11: Coefficients of demographic variables from Probit regression of Democratic/Republican party affiliation using demographic variables common to American National Election Survey (ANES) and Michigan Survey of Consumers (MSC) cross-sectional data. 2020 Presidential election, with sample period from Jan 2019 to Dec 2022.

2024 (MSC)	Democrat		Republican	
Age	-0.0005	(0.0094)	0.0130	(0.0095)
Age squared	0.0001	(0.0001)	-0.0001	(0.0001)
Female	0.2756***	(0.0425)	-0.2665***	(0.0428)
HS or less	-0.2831**	(0.1355)	0.0082	(0.1256)
Some College	0.1265**	(0.0645)	-0.0890	(0.0636)
BA Degree	0.5004***	(0.0573)	-0.3255***	(0.0573)
Post BA	0.7518***	(0.063)	-0.6077***	(0.0642)
Married	-0.2545***	(0.048)	0.2776***	(0.0483)
3+ children	-0.3849***	(0.0902)	0.3302***	(0.0858)
Own Residence	-0.0252	(0.0538)	0.0595	(0.0538)
Income Mid	0.0505	(0.0562)	0.0509	(0.0563)
Income High	0.1994***	(0.063)	-0.0684	(0.0636)
Constant	-0.4332*	(0.2273)	-0.7057***	(0.2293)
Obs	3,769		3,769	
R^2	0.078		0.059	

Table A.12: Coefficients of demographic variables from Probit regression of Democratic/Republican party affiliation using demographic variables common to American National Election Survey (ANES) and Michigan Survey of Consumers (MSC) cross-sectional data. 2024 Presidential election, with sample period from Jan 2023 to November 2025.

2016 (SCE)	Democrat		Republican	
Age	0.0145	(0.0103)	0.0102	(0.0105)
Age squared	-0.0002	(0.0001)	0.0000	(0.0001)
Female	0.2993	(0.0531)***	-0.2213	(0.0542)***
Black	1.2749	(0.1125)***	-1.7102	(0.1602)***
Asian	0.1588	(0.1452)	-0.3948	(0.1555)**
Native American	0.6905	(0.4262)	-0.8248	(0.4566)*
Hispanic	0.8564	(0.0967)***	-0.8534	(0.1045)***
Multiracial	0.4035	(0.1361)***	-0.3351	(0.1385)**
HS or less	0.0402	(0.1418)	0.0796	(0.1462)
Some College	0.0838	(0.0855)	0.0066	(0.0862)
Assoc. Degree	-0.0424	(0.0945)	0.1141	(0.0942)
BA Degree	0.3179	(0.0828)***	-0.2392	(0.0834)***
Post BA	0.7240	(0.0923)***	-0.5719	(0.0937)***
Temporarily Laid Off	-0.1638	(0.2911)	-0.0097	(0.2975)
Unemployed	-0.0232	(0.125)	-0.0497	(0.1321)
Retired	0.1126	(0.0972)	-0.1377	(0.0973)
Permanently Disabled	-0.0060	(0.1459)	-0.1139	(0.1555)
Homemaker	-0.4402	(0.1265)***	0.3452	(0.1226)***
Student	0.3137	(0.1648)*	-0.0479	(0.1811)
Married	-0.0598	(0.0652)	0.1623	(0.0663)**
Spouse: Temporarily Laid Off	-0.0387	(0.4342)	-0.2555	(0.4473)
Spouse: Unemployed	-0.1291	(0.1644)	0.0357	(0.1717)
Spouse: Retired	0.1450	(0.0972)	-0.1696	(0.0974)*
Spouse: Permanently Disabled	-0.0244	(0.1865)	-0.0431	(0.1892)
Spouse: Homemaker	-0.1322	(0.1153)	-0.0259	(0.1119)
Spouse: Student	-0.1030	(0.2229)	0.2852	(0.226)
Resided 3+yrs in Current Address	0.0337	(0.0614)	-0.0262	(0.063)
Own Residence	-0.2412	(0.0707)***	0.2557	(0.0726)***
Other Residence	-0.1647	(0.1118)	0.1845	(0.1166)
3+ children	-0.2945	(0.1156)**	0.2287	(0.114)**
Income Mid	-0.0583	(0.0663)	0.1490	(0.0679)**
Income High	0.0514	(0.0813)	-0.0041	(0.0829)
Constant	-0.6975	(0.2719)**	-0.7665	(0.2805)***
Obs.	2,810		2,810	
R^2	0.133		0.150	

Table A.13: Coefficients of demographic variables from Probit regression of Democratic/Republican party affiliation using demographic variables common to American National Election Survey (ANES) and Survey of Consumer Expectations (SCE) cross-sectional data. 2016 Presidential election, with sample period from Jan 2015 to Dec 2018.

2020 (SCE)	Democrat		Republican	
Age	-0.0227	(0.0172)	0.0442	(0.018)**
Age squared	0.0002	(0.0002)	-0.0004	(0.0002)**
Female	0.0601	(0.0638)	-0.0402	(0.0658)
Black	0.6326	(0.1565)***	-1.4077	(0.2342)***
Asian	-0.0766	(0.1598)	-0.2219	(0.17)
Native American	0.0929	(0.2461)	-0.1839	(0.2529)
Hispanic	0.3342	(0.1154)***	-0.5155	(0.1222)***
Multiracial	0.2803	(0.1741)	-0.2922	(0.1827)
HS or less	-0.2591	(0.2091)	0.0311	(0.1959)
Some College	0.0941	(0.1142)	0.0438	(0.1122)
Assoc. Degree	0.0838	(0.1217)	0.0134	(0.119)
BA Degree	0.4346	(0.1085)***	-0.2775	(0.1078)**
Post BA	0.6131	(0.1116)***	-0.4876	(0.1127)***
Temporarily Laid Off	0.1595	(0.2067)	-0.0137	(0.2153)
Unemployed	-0.5285	(0.4016)	0.4455	(0.368)
Retired	-0.0837	(0.1216)	0.0131	(0.1213)
Permanently Disabled	0.1919	(0.2049)	-0.0173	(0.2121)
Homemaker	-0.2848	(0.1323)**	0.1971	(0.1285)
Student	0.8045	(0.6288)	-1.2620	(0.9386)
Married	-0.0755	(0.0949)	0.2073	(0.1019)**
Spouse: Temporarily Laid Off	-0.0487	(0.1969)	-0.1042	(0.2019)
Spouse: Unemployed	-0.1502	(0.1454)	-0.0050	(0.1506)
Spouse: Retired	0.1435	(0.1118)	-0.0229	(0.1124)
Spouse: Permanently Disabled	-0.0982	(0.2061)	-0.0734	(0.2109)
Spouse: Homemaker	-0.1372	(0.1357)	0.1289	(0.1336)
Spouse: Student	0.0983	(0.3676)	0.3276	(0.3777)
Resided 3+yrs in Current Address	0.0220	(0.0785)	0.1038	(0.0818)
Own Residence	-0.1786	(0.0922)*	0.1031	(0.0978)
Other Residence	-0.3184	(0.2313)	0.0924	(0.2269)
3+ children	-0.2639	(0.108)**	0.2613	(0.1067)**
Income Mid	0.1018	(0.0948)	0.0946	(0.0954)
Income High	0.1572	(0.1007)	-0.0506	(0.1024)
Constant	0.2748	(0.4261)	-1.5831	(0.4486)***
Obs.	1,941		1,941	
R^2	0.083		0.110	

Table A.14: Coefficients of demographic variables from Probit regression of Democratic/Republican party affiliation using demographic variables common to American National Election Survey (ANES) and Survey of Consumer Expectations (SCE) cross-sectional data. 2020 Presidential election, with sample period from Jan 2019 to Dec 2022.

2024 (SCE)	Democrat		Republican	
Age	0.0003	(0.0108)	0.0100	(0.0109)
Age squared	0.0000	(0.0001)	-0.0001	(0.0001)
Female	0.2831	(0.0454)***	-0.2914	(0.046)***
Black	1.0765	(0.0959)***	-1.2291	(0.1126)***
Asian	-0.2156	(0.1197)*	-0.1418	(0.1236)
Native American	-0.1203	(0.3019)	-0.0859	(0.2874)
Hispanic	0.2629	(0.0771)***	-0.5255	(0.0798)***
Multiracial	0.1342	(0.1178)	-0.2106	(0.1192)*
HS or less	-0.3094	(0.1464)**	0.1000	(0.1361)
Some College	0.2155	(0.0785)***	-0.1065	(0.0769)
Assoc. Degree	0.1770	(0.0837)**	-0.0614	(0.0816)
BA Degree	0.6500	(0.0735)***	-0.4084	(0.0721)***
Post BA	0.9023	(0.0793)***	-0.6909	(0.0791)***
Temporarily Laid Off	0.6208	(0.2517)**	-0.3607	(0.2613)
Unemployed	0.1062	(0.0672)	-0.1244	(0.0679)*
Retired	0.0821	(0.0766)	-0.0609	(0.0771)
Permanently Disabled	0.1113	(0.1031)	-0.1716	(0.1041)*
Homemaker	-0.1231	(0.0764)	0.0822	(0.077)
Student	0.2561	(0.1375)*	-0.3679	(0.1474)**
Married	-0.2026	(0.0562)***	0.2283	(0.057)***
Spouse: Temporarily Laid Off	-0.0681	(0.3081)	-0.0911	(0.3224)
Spouse: Unemployed	0.0415	(0.1431)	-0.0937	(0.1419)
Spouse: Retired	0.1113	(0.0797)	-0.0969	(0.0801)
Spouse: Permanently Disabled	0.0872	(0.1461)	-0.1895	(0.1456)
Spouse: Homemaker	0.0116	(0.1047)	-0.0258	(0.1019)
Spouse: Student	0.2801	(0.2447)	-0.1393	(0.2479)
Resided 3+yrs in Current Address	-0.0343	(0.0496)	0.0674	(0.0502)
Own Residence	-0.0353	(0.0703)	0.1114	(0.0712)
Other Residence	-0.0958	(0.0601)	0.1486	(0.0612)**
3+ children	-0.3970	(0.0939)***	0.3627	(0.0897)***
Income Mid	0.1140	(0.0597)*	-0.0362	(0.0601)
Income High	0.3032	(0.069)***	-0.2261	(0.0699)***
Constant	-0.6291	(0.2779)**	-0.4398	(0.2815)
Obs.	3,799		3,799	
R^2	0.127		0.115	

Table A.15: Coefficients of demographic variables from Probit regression of Democratic/Republican party affiliation using demographic variables common to American National Election Survey (ANES) and Survey of Consumer Expectations (SCE) cross-sectional data. 2024 Presidential election, with sample period from Jan 2023 to April 2025.

MSC imputations (%)	2004	2008	2012	2016	2020	2024
True Positives	18.74	35.84	32.02	34.55	40.70	40.87
True Negatives	35.02	20.73	23.13	25.56	17.08	19.79
False Positives	11.64	27.23	22.93	23.27	31.93	26.05
False Negatives	34.60	16.20	21.93	16.62	10.29	13.29
Correct imputation	53.76	56.57	55.15	60.11	57.78	60.66
Regions-only imputation	48.51	50.39	53.94	47.88	47.85	50.71
Obs.	971	5,759	1,496	10,707	18,170	10,571

Table A.16: Accuracy of imputation method using MSC data. All numbers are in percentages of the total sample for each Presidential election period (2004, 2008, 2012, 2016, 2020, and 2024). True positives: Imputation method correctly identifies the respondent as a Democrat. True negatives: Imputation method correctly identifies the respondent as a Republican. False positives: The imputation method falsely identifies the respondent as a Democrat. False negatives: The imputation method falsely identifies the respondent as a Republican. Regions-only imputation: accuracy of imputation given only U.S. Census region dummies (Northeast, Midwest, South, and West).

MSC imputations (%)	2004	2008	2012	2016	2020	2024
True Positives	22.86	34.40	33.56	30.41	34.25	36.05
True Negatives	27.70	20.92	21.19	24.69	20.80	21.91
False Positives	18.95	27.04	24.87	24.14	28.21	23.93
False Negatives	30.48	17.64	20.39	20.75	16.73	18.11
Correct imputation	50.57	55.32	54.75	55.10	55.06	57.96
Obs.	971	5,759	1,496	10,707	18,170	10,571

Table A.17: Accuracy of imputation method using MSC data, random forest model.

MSC imputations (%)	2004	2008	2012	2016	2020	2024
True Positives	20.70	34.71	32.82	32.10	37.91	37.42
True Negatives	33.99	22.24	22.86	27.26	19.50	21.88
False Positives	12.67	25.72	23.20	21.57	29.52	23.96
False Negatives	32.65	17.33	21.12	19.06	13.07	16.73
Correct imputation	54.69	56.95	55.68	59.36	57.41	59.30
Obs.	971	5,759	1,496	10,707	18,170	10,571

Table A.18: Accuracy of imputation method using MSC data, XGBoost model.

Variable description	Horizon	Macro/Micro	Survey	Election 1980		Election 1984		Election 1988		Election 1992		Election 1996		Election 2000	
				Switch	Bias	Switch	Bias	Switch	Bias	Switch	Bias	Switch	Bias	Switch	Bias
Labour Market															
Prob of higher unemployment rate	12mo	Macro	MSC	-0.092***	0	0.055***	-0.110**	0.025**	-0.111***	0.028**	0	0	0	-0.058***	0
Prob of higher nominal income	12mo	Micro	MSC	0	0	-0.459*	2.695**	0	-2.342***	0	-2.872*	-0.479**	0	0	0
Higher vs lower real household income	12mo	Micro/Macro	MSC	0.026***	0.059**	-0.025**	0.143***	0	-0.092***	0	0	0	0	0	0.048*
Better or worse personal financial situation (future)	12mo	Micro	MSC	0.030***	-0.070***	-0.029***	0	0	0.078***	0	-0.150**	0	0	0	0.075***
Micro Inflation															
Expected Increase in gas prices	12mo	Micro/Macro	MSC	0	0	0	-6.908***	0	0	0.000***	-12.692*	0	0	0	0
Expected Increase in gas prices	5y	Micro/Macro	MSC	0	0	0	-29.870**	0	0	0	-53.522***	0	9.637***	0	5.556*
Macro Variables															
Expected inflation rate	12mo	Macro	MSC	-0.602**	0	0	0	0	-0.778**	0	-2.738***	-0.172*	0	-0.216**	0.412*
Expected inflation rate	5y	Macro	MSC	-0.997*	0	0	0	0	0	0	0	0	0.508**	-0.239***	0.349*
Interest rates rise/fall	12mo	Macro	MSC	-0.051***	0.087***	0	0.176***	-0.069***	0.088***	0.035***	-0.186**	-0.029**	0	0	0
Business conditions improve/decline	12mo	Macro	MSC	0.061***	0	-0.049***	0.171***	0	0.106***	-0.039***	0.343***	0.036***	0.075***	0.061***	0
Business financial conditions improve/decline	12mo	Macro	MSC	0	-0.091***	-0.032**	0.167**	0	0.141***	-0.040**	0	0.046***	-0.063**	0.059***	0
Index of Consumer Sentiment	12mo	Macro	MSC	2.624***	-9.957***	-4.863***	13.399**	0	9.037***	-4.575***	0	0	-7.635***	5.092***	4.558**
Index of Current Economic Conditions	12mo	Macro	MSC	0	-7.536**	-6.314***	0	0	8.457**	-3.835**	-27.248**	0	-10.896***	2.616*	0
Index of Consumer Expectations	12mo	Macro	MSC	3.906***	-11.511***	-3.931***	15.683**	0	7.253*	-5.051***	0	3.643**	-6.338**	6.382***	5.316*
Macro Sentiment Index	12mo	Macro	MSC	4.798**	-14.841***	-5.395*	27.985**	0	0	-10.752***	0	9.399***	-14.000**	13.924***	0
Micro Sentiment Index	12mo	Macro	MSC	1.939*	-8.628***	-5.720***	9.976*	0	9.620***	-2.661**	-25.888***	-1.953*	-7.061***	2.307**	6.234***

Table A.19: Comparison of Expectation Variables throughout Elections 1980-2000

Variable description	Horizon	Macro/Micro	Survey	Election 2004		Election 2008		Election 2012		Election 2016		Election 2020		Election 2024	
				Switch	Bias	Switch	Bias	Switch	Bias	Switch	Bias	Switch	Bias	Switch	Bias
Labour Market															
Prob of higher unemployment rate	12mo	Macro	MSC	0	-0.295***	0	0	-0.026**	0.221***	-0.146***	0	-0.141***	0.046**	-0.158***	0
Prob of higher unemployment rate	12mo	Macro	SCE							-7.390***	0	-7.740***	0	-11.232***	0
Prob of losing job	5y	Micro	MSC	0	-0.073***	0.015**	-0.053***	0	0	-0.034***	-0.032**	-0.020***	0	-0.030***	0
Perceived job separation prob	12mo	Micro	SCE							0	-3.750***	0	0	-2.435*	0
Perceived job finding prob	3mo	Micro	SCE							3.414**	0	0	0	0	-9.428**
Prob of quitting	12mo	Micro	SCE							0	-5.084*	0	0	0	-8.627***
Higher or lower personal wage rate	12mo	Micro	SCE							0	-0.122***	-0.057**	0	0	0
Prob of higher nominal income	12mo	Micro	MSC	0	3.381***	0	0	0	0	0.571***	0.645*	1.069***	0	1.063***	1.260**
Percentage of real wage growth	12mo	Micro	SCE							0	-0.895**	0	0	0	0
Higher vs lower real household income	12mo	Micro/Macro	MSC	0	0.241***	0.036***	0.078***	0	0	0.064***	0	0.075***	0	0.098***	-0.056*
Better or worse personal financial situation (future)	12mo	Micro	MSC	0	0.285***	0	0.032*	0	0	0.067***	0.033*	0.117***	0	0.141***	0
Higher or lower nominal household income	12mo	Micro	SCE							-0.088***	-0.133***	-0.067***	0.363**	-0.113***	0
Percentage of real household income growth	12mo	Micro/Macro	SCE							0.816*	0	0	0	1.428*	0
Household Spending															
Higher or lower spending	12mo	Micro	SCE							0	0	0	0	0	0
Percentage of spending growth	12mo	Micro	SCE							0	0	0	4.817**	-2.074***	0
Micro Inflation															
Expected Increase in gas prices	12mo	Micro/Macro	MSC	0	0	0	0	0	-23.177***	-1.646*	3.695**	-7.552***	6.796**	0	0
Expected Increase in gas prices	5y	Micro/Macro	MSC	0	0	0	22.062***	0	-80.988***	-5.558**	18.544***	-12.493***	16.049***	-5.153*	25.852***
Expected Increase in gas prices	12mo	Micro/Macro	SCE							-1.816***	-1.045***	-2.595***	0	-4.046***	0
Expected Increase in food prices	12mo	Micro/Macro	SCE							-0.764***	-0.856***	-1.538***	0	-4.572***	0
Expected Increase in medical prices	12mo	Micro/Macro	SCE							-2.405***	0	-0.833*	0	-3.501***	0
Expected increase in college prices	12mo	Micro/Macro	SCE							-0.869**	0	0	0	-2.516***	0
Expected increase in rent	12mo	Micro/Macro	SCE							0	0	-1.135***	0	-3.249***	0
Expected increase in gold	12mo	Micro/Macro	SCE							0	0	-0.715*	0	-2.111***	0
Macro Variables															
Expected inflation rate	12mo	Macro	MSC	0	0	-0.305**	0	0	1.778***	-0.438***	0.492***	-1.276***	0.625**	-1.389***	0
Expected inflation rate	5y	Macro	MSC	0	0	-0.280***	0	0	0	-0.245***	0.309**	-0.196**	0.568***	0	0
Inflation or deflation	12mo	Macro	SCE							0.029***	-0.114***	0.022**	0	0.138***	0
Expected inflation rate	12mo	Macro	SCE							-1.361***	0	-1.416***	0	-4.675***	0
Higher or lower house prices	12mo	Macro	SCE							0	-0.088**	-0.101***	0.339**	0.093**	0
Expected house price change	12mo	Macro	SCE							0	0	0	-3.282***	-2.074***	0
Prob of higher interest rate	12mo	Macro	SCE							3.134***	0	0	0	-1.761*	-7.163***
Interest rates rise/fall	12mo	Macro	MSC	-0.021**	0	-0.062***	0	-0.025***	0	0	0	0	0	-0.078***	0
Prob of higher stock market	12mo	Macro	SCE							4.719***	0	3.327***	0	3.658***	-11.442***
Business conditions improve/decline	12mo	Macro	MSC	0	0.214***	0.019*	-0.051**	0.035***	0	0.160***	0	0.173***	-0.042*	0.217***	0
Business financial conditions improve/decline	12mo	Macro	MSC	0	0.425***	0.080***	0	0	-0.226***	0.218***	0	0.225***	0	0.197***	-0.072*
Index of Consumer Sentiment	12mo	Macro	MSC	0	38.103***	5.258***	0	0	-20.209***	18.390***	0	21.912***	0	19.701***	-6.209*
Index of Current Economic Conditions	12mo	Macro	MSC	0	21.122***	0	0	0	-17.236**	8.973***	4.283*	14.778***	5.839**	12.893***	0
Index of Consumer Expectations	12mo	Macro	MSC	0	49.011***	7.348***	0	0	-22.117***	24.438***	0	26.494***	0	24.073***	-9.536**
Macro Sentiment Index	12mo	Macro	MSC	0	79.324***	16.968***	0	0	-45.897***	47.601***	0	46.354***	0	39.518***	-18.490***
Micro Sentiment Index	12mo	Macro	MSC	0	28.600***	0	0	0	-11.859*	9.406***	4.521**	15.852***	5.489**	15.815***	0

Table A.20: Comparison of Expectation Variables throughout Elections 2004-2024

A.4 Additional results on real versus real wage/income growth expectations

In Figures 1 and 2 of the Introduction, we have shown that partisan expectations of macroeconomic variables shift by more than those of microeconomic variables or personal finances shortly around U.S. presidential elections. Thus, in this section, we further demonstrate our empirical findings in several steps.

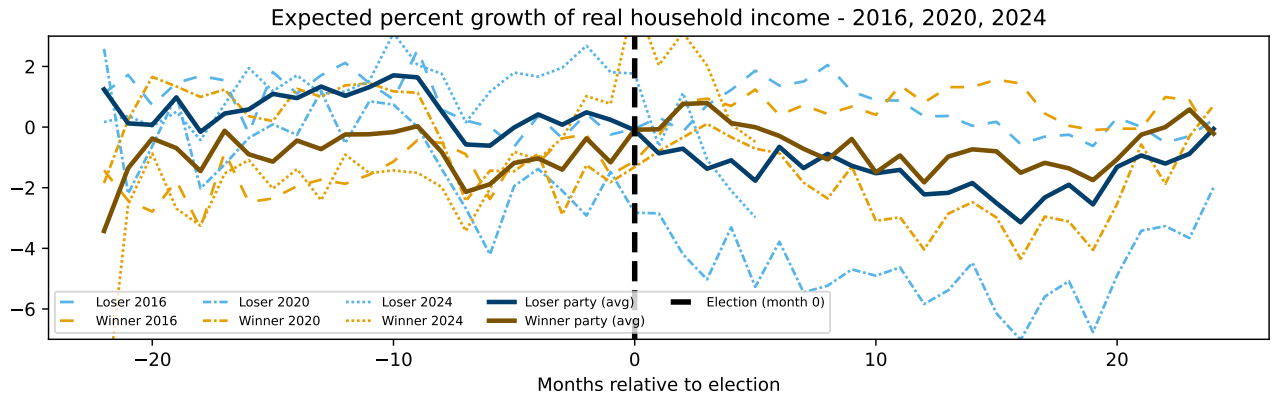
We first summarize our findings on household expectations of real income growth in the next 12 months, with survey data from the SCE. We find partisan differences to be punctuated shortly after presidential elections, as seen in Figure A.1a. The expectations on real household income growth for respondents supporting the winning party are consistently greater than those of the losing party, as averaged across the 2016, 2020, and 2024 elections.

A similar qualitative pattern appears for respondents’ own real wage growth expectations in Subfigures A.2b. This difference is much less visually apparent in *nominal* household income expectations (Figure A.2 in the Appendix), which implies that these changes in partisan expectations over time are primarily driven by partisan inflation expectations.

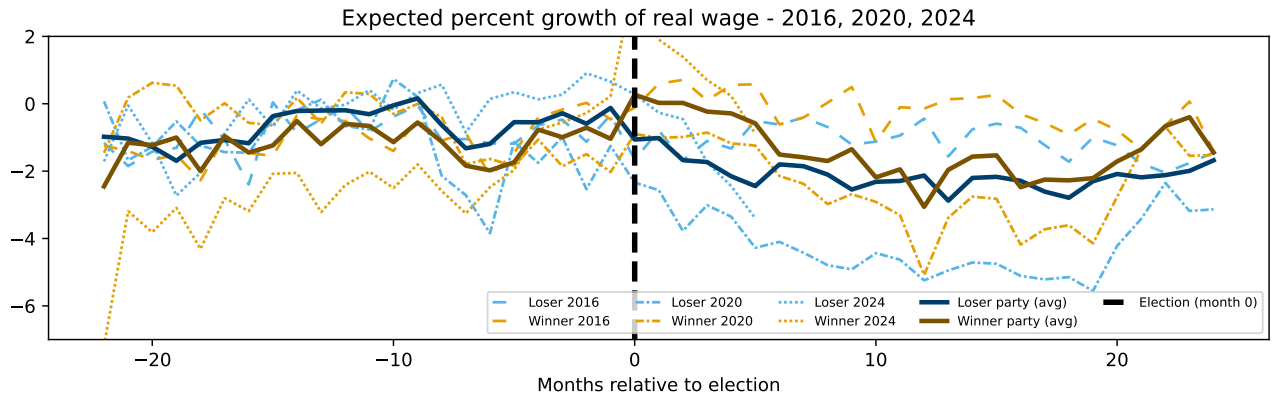
Interestingly, we do not find sharp partisan differences around presidential elections for nominal wage growth, as shown in Figure A.2b. Note that here we treat *nominal* wage and income expectations as *micro* expectations as they are elicited regarding the respondents’ own income/wage changes. This differs from the inflation expectations which necessarily regard the overall price level. Households are more likely to have direct references on their own income/wage growth as opposed to aggregate inflation.²⁶ The results above again confirm that micro expectations exhibit fewer partisan switches than those regarding the aggregate inflation rate.

²⁶Our interpretation is related to, but distinct from, the “money illusion”, that individuals are not able to perfectly disentangle real changes from nominal counterparts.

Figure A.1: Income/Earning Expectations by Inferred Partisanship, 2016-2024.



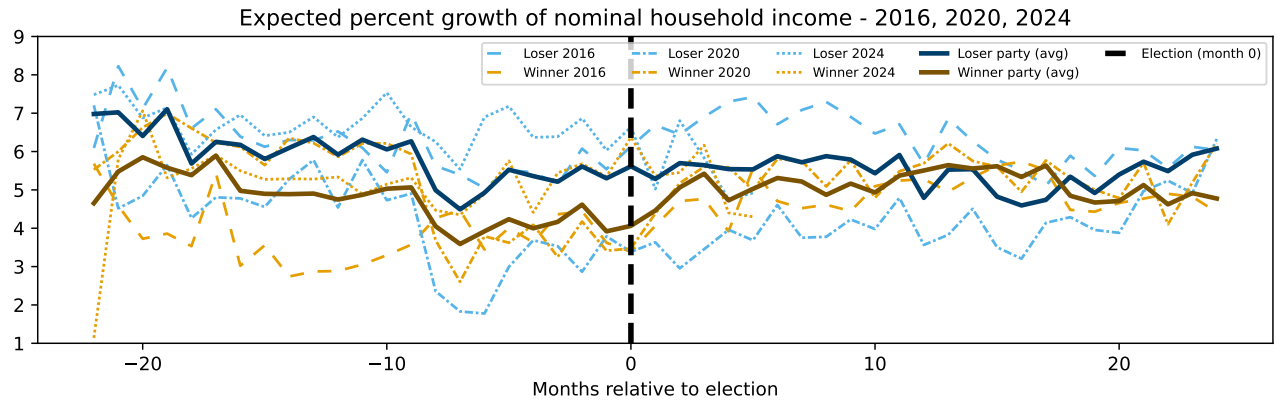
(a) Real household income growth rate in percentage points, next 12 months, 2016-2024



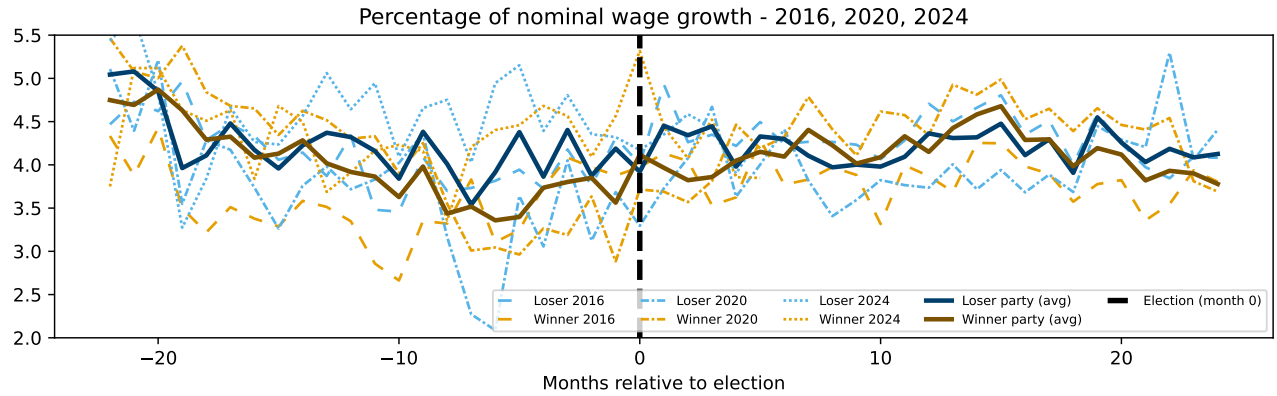
(b) Real wage growth rate in percentage points, next 12 months, 2016-2024

Notes: These tables are generated using data from the 2016, 2020, and 2024 American National Election Survey (ANES) and the Survey of Consumer Expectations (SCE) from the calendar year before and after the election. Orange lines indicate the mean responses of respondents affiliated with the winning party, blue lines indicate those respondents affiliated with the losing one.

Figure A.2: **Nominal** Income/Earning Expectations by Inferred Partisanship.



(a) Nominal household income growth rate in percentage points, next 12 months



(b) Nominal wage growth rate in percentage points, next 12 months

Notes: These tables are generated using data from the 2016, 2020, and 2024 American National Election Survey (ANES) and the Survey of Consumer Expectations (SCE) from the calendar year before and after the election. Orange lines indicate the mean responses of respondents affiliated with the winning party, blue lines indicate those respondents affiliated with the losing one.

Policy beliefs, all election cycles	(1) Welfare Benefits				
	Change	$\times \text{Dem}_i$	$\times \text{WinRoot}_{it}$	$\times \text{PostElect}_t$	$\times \text{WinRoot}_{it} \times \text{PostElect}_t$
Prob of higher unemployment rate	0	***	0	0	***
Perceived job separation prob	***	***	***	0	***
Percentage of real wage growth	0	***	***	0	0
Percentage of real household income growth	0	0	0	0	0
Expected inflation rate	0	0	0	0	0
Expected Increase in gas prices	+	***	0	0	0
Expected Increase in food prices	0	**	0	+	0
	(2) Unemployment Benefits				
	Change	$\times \text{Dem}_i$	$\times \text{WinRoot}_{it}$	$\times \text{PostElect}_t$	$\times \text{WinRoot}_{it} \times \text{PostElect}_t$
Prob of higher unemployment rate	***	0	**	0	***
Perceived job separation prob	***	***	***	0	**
Percentage of real wage growth	0	**	0	0	0
Percentage of real household income growth	0	0	0	0	0
Expected inflation rate	0	0	***	0	0
Expected Increase in gas prices	0	*	0	0	0
Expected Increase in food prices	0	0	0	0	0
	(3) Payroll Tax				
	Change	$\times \text{Dem}_i$	$\times \text{WinRoot}_{it}$	$\times \text{PostElect}_t$	$\times \text{WinRoot}_{it} \times \text{PostElect}_t$
Prob of higher unemployment rate	***	***	**	0	+
Perceived job separation prob	0	0	0	0	0
Percentage of real wage growth	0	0	0	0	***
Percentage of real household income growth	0	**	***	0	***
Expected inflation rate	***	***	***	0	***
Expected Increase in gas prices	***	0	0	0	***
Expected Increase in food prices	***	***	***	0	0
	(4) Capital Gains Tax				
	Change	$\times \text{Dem}_i$	$\times \text{WinRoot}_{it}$	$\times \text{PostElect}_t$	$\times \text{WinRoot}_{it} \times \text{PostElect}_t$
Prob of higher unemployment rate	0	***	0	0	***
Perceived job separation prob	0	0	0	0	0
Percentage of real wage growth	0	0	0	0	0
Percentage of real household income growth	0	0	*	***	+
Expected inflation rate	***	0	**	0	0
Expected Increase in gas prices	***	***	0	0	0
Expected Increase in food prices	0	0	0	0	0
	(5) Income Tax				
	Change	$\times \text{Dem}_i$	$\times \text{WinRoot}_{it}$	$\times \text{PostElect}_t$	$\times \text{WinRoot}_{it} \times \text{PostElect}_t$
Prob of higher unemployment rate	***	***	*	0	***
Perceived job separation prob	0	***	0	***	*
Percentage of real wage growth	0	0	0	*	0
Percentage of real household income growth	0	***	***	0	**
Expected inflation rate	***	***	0	0	0
Expected Increase in gas prices	***	0	0	0	0
Expected Increase in food prices	***	***	0	0	0
	(6) Income Tax for Highest Income Bracket				
	Change	$\times \text{Dem}_i$	$\times \text{WinRoot}_{it}$	$\times \text{PostElect}_t$	$\times \text{WinRoot}_{it} \times \text{PostElect}_t$
Prob of higher unemployment rate	0	0	**	0	***
Perceived job separation prob	0	0	0	0	0
Percentage of real wage growth	0	***	***	0	0
Percentage of real household income growth	0	0	0	**	***
Expected inflation rate	0	***	0	0	0
Expected Increase in gas prices	0	**	***	0	0
Expected Increase in food prices	***	***	***	***	0

Table A.21: Signs and significance levels of coefficients in regressions of beliefs on expected public policy direction changes. Results are pooled across the 2016, 2020, and 2024 election cycles. For each policy panel, Change is the policy-direction variable; the other columns interact Change with the listed indicator. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

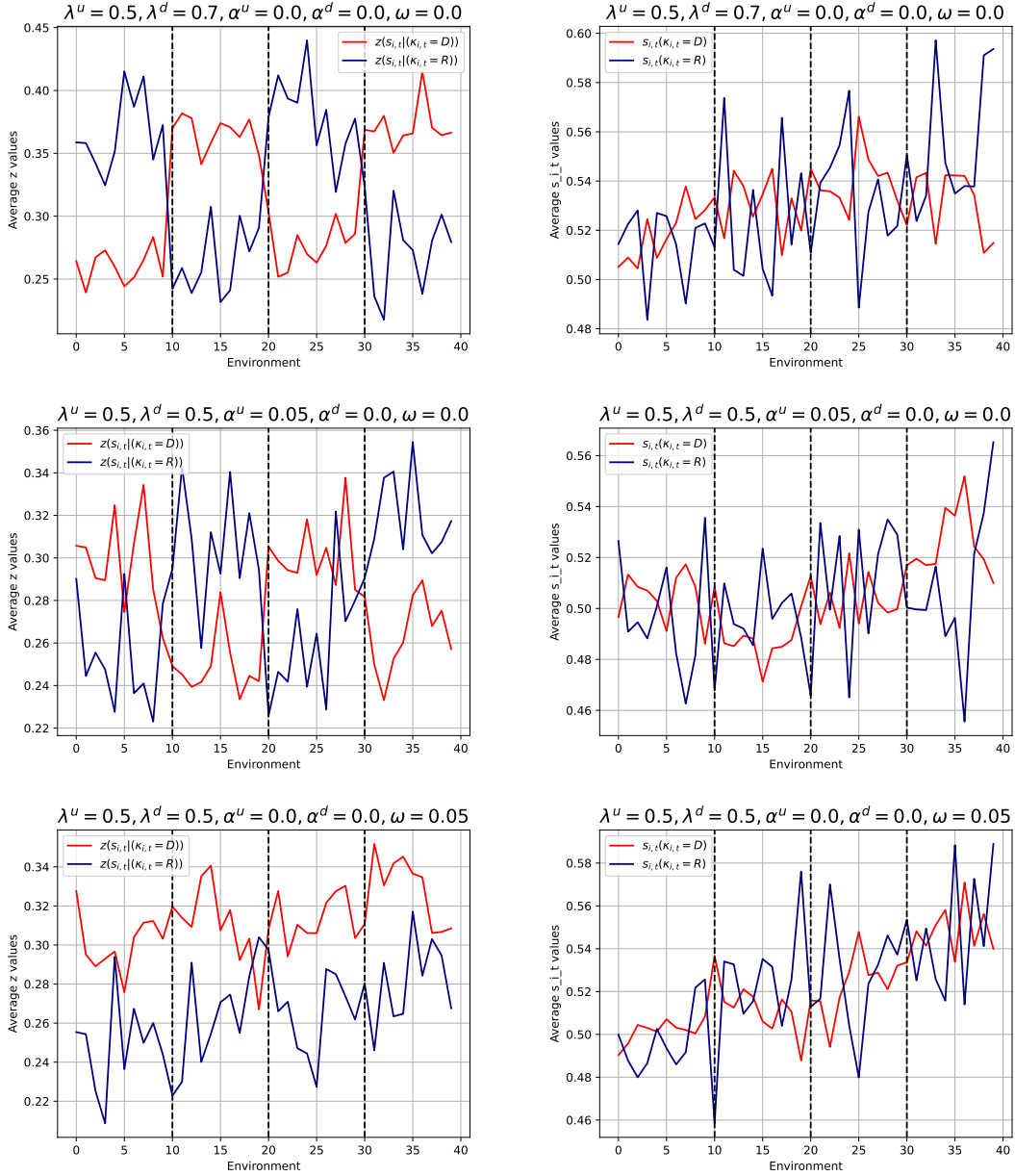
	Welfare Benefits		Unemployment Benefits		Payroll Tax		Capital Gains Tax		Income Tax		Income Tax, High Incomes	
	Direction	Effect	Direction	Effect	Direction	Effect	Direction	Effect	Direction	Effect	Direction	Effect
	Welfare Benefits		Unemployment Benefits		Payroll Tax		Capital Gains Tax		Income Tax		Income Tax, High Incomes	
	Direction	Effect	Direction	Effect	Direction	Effect	Direction	Effect	Direction	Effect	Direction	Effect
Prob of higher unemployment rate	0	***	0	***	0	***	0	***	0	***	***	***
Perceived job separation prob	0	***	***	***	*	**	0	***	***	***	***	***
Perceived job finding prob	*	***	0	***	***	***	0	***	***	***	0	***
Prob of hiring more workers	0	+	0	0	0	**	***	***	***	***	0	***
Prob of quitting	0	**	0	0	0	***	+	0	*	***	0	0
Prob of finding job in 12 months	0	0	***	0	0	***	0	0	0	0	*	0
Prob of finding job in 3 months	0	0	0	0	0	*	0	0	0	0	0	0
Prob of job search in 12 months	**	**	**	0	0	***	0	***	***	+	0	0
Prob of job search in 3 months	***	0	***	0	0	0	0	0	***	***	*	0
Higher/lower personal wage rate	0	***	0	***	**	***	***	***	0	***	0	***
Percentage of real wage growth	0	***	**	***	0	***	0	***	0	***	0	***
Higher/lower real household income	***	***	0	***	***	***	***	***	0	***	0	***
Percentage of real household income growth	0	***	0	+	***	+	***	***	0	***	0	0
Interest rates rise/fall	0	0	0	0	0	***	0	0	0	0	0	***
Prob of higher stock market	0	***	*	***	***	***	*	***	***	***	0	***
Inflation/deflation	**	0	0	***	**	+	***	0	0	***	0	+
Expected inflation rate	+	***	+	***	***	**	***	***	***	0	0	***
Easier to get credit now	0	***	0	***	***	***	0	***	0	***	0	***
Easier to get credit in future	0	***	0	***	0	***	0	***	0	***	0	***
Higher/lower house prices	0	0	0	0	0	***	**	0	***	0	0	0
Expected house price change	+	0	0	**	0	**	***	0	0	0	0	0
Expected probability of spending growth	***	**	***	***	***	0	***	0	***	0	***	***
Expected spending growth rate	***	***	0	***	***	0	***	**	***	**	0	***
Expected Increase in gas prices	0	***	0	***	***	***	***	***	***	***	0	***
Expected Increase in food prices	0	***	0	***	0	***	0	***	***	***	0	***
Expected Increase in medical prices	0	***	0	***	+	***	0	***	0	***	**	***
Expected increase in college prices	0	***	0	***	+	***	0	***	0	***	***	***
Expected increase in rent	0	***	0	***	0	***	***	***	***	***	0	***

Table A.22: Signs and significance levels of coefficients in regressions of policy beliefs on selected macro/micro expectations variables. Data from the SCE's Public Policy Survey (PPS) sub-module. For each type of public policy (columns), "Direction" stands for the direction of the policy change, on a scale of -1 to 1 (-1 for reduce/decrease, 1 for expand/increase), and "Effect" stands for the SCE PPS respondent's expectations of the effects of the respondent's expected policy change on their own households, on a scale from -1 to 1 (-1 for negative, 1 for positive). Separate regression results for each type of public policy on each type of micro/macro beliefs (rows). *, **, and *** represent significance of the coefficient at the 10%, 5%, and 1% levels.

B Additional Model Results

Figure A.3 plots the simulated average beliefs of respondents from each party over a sample period covering three assumed switches in election outcomes.

Figure A.3: Model simulation with different parameter values



C Details of the Bayesian Estimation

Table A.23 reports the prior distribution used for each parameter in the estimation.

Table A.23: Priors

Parameter	Prior	Rationale
p	Beta(2, 2)	Symmetric; weakly informative around 0.5
e_1, e_0	Beta(1, 4)	Mean 0.2; mislabeling uncommon but possible
$\alpha, \delta, \lambda_{\text{up}}, \lambda_{\text{down}}, \omega_1$	$\mathcal{N}(0, 2)$	Weakly informative; variables are standardized
σ	HalfNormal(2)	Positive support; weakly informative

All continuous expectation variables ($s_{i,t}, \psi_{i,t}$) are standardized prior to estimation, so Normal(0, 2) priors are weakly informative on a unit-variance scale.

Posterior Sampling: NUTS. Posterior inference is carried out via Markov Chain Monte Carlo using the No-U-Turn Sampler (NUTS), a self-tuning variant of Hamiltonian Monte Carlo (HMC), implemented in PyMC (Abril-Pla et al., 2023).

Algorithm sketch. At each iteration, NUTS augments the parameter vector $\theta \in \mathbb{R}^d$ with a momentum variable $r \sim \mathcal{N}(0, M)$, where M is a diagonal mass matrix adapted during warm-up. It then simulates Hamiltonian dynamics

$$H(\theta, r) = -\log \pi(\theta \mid \text{data}) + \frac{1}{2} r^\top M^{-1} r, \quad (14)$$

by leapfrog integration with step size ε . Unlike standard HMC, which requires the user to specify the number of leapfrog steps, NUTS grows a binary tree of trajectory proposals forward and backward in time, doubling the path length at each level, and stops when the trajectory begins to double back on itself (the “U-turn” criterion). A proposal is drawn uniformly from the resulting set of tree leaves via slice sampling, and accepted with a Metropolis correction. Because the dynamics are nearly energy-preserving, acceptance rates are high, and the sampler moves efficiently through the posterior.

Tuning. During the warm-up phase, PyMC adapts ε and the diagonal of M using a dual-averaging scheme targeting an acceptance rate of 0.95. This high target acceptance rate is chosen to encourage small step sizes and careful exploration, which matters given the mixture structure of the likelihood: the marginalization over k_i^* can produce multimodal or funnel-shaped posteriors for the mislabeling parameters.

Sampler settings. See Table A.24.

Table A.24: Sampler setting

Setting	Value
Chains	4 (run in parallel)
Warm-up (tuning) draws	1,500
Post-warm-up draws per chain	1,500
Target acceptance rate	0.95
Posterior subsample for decomposition	500

C.1 Convergence Diagnostics

Convergence is monitored using the split $\hat{R}$ statistic, which compares within-chain to between-chain variance. Values near 1 indicate that all chains have mixed to the same distribution. We flag any model-year-variable combination for which $\max_{\theta} \hat{R} > 1.01$ and re-estimate it with an extended budget (2,000 draws, 2,000 tuning steps, target acceptance rate 0.96). We additionally report bulk and tail effective sample sizes (ESS) to verify that the tails of the posterior are adequately explored.

C.2 Estimation results based on imputed likelihood

See Figure [A.4](#) and [A.5](#).

Figure [A.5](#) evaluates the contributions of all partisan mechanisms to the observed macroeconomic belief dispersion, based on the frequentist parameter estimates from MLE. We measure such importance by the explanatory power of our factor-model estimates, as measured by R^2 from a linear regression of observed $z_{i,t}$ on $\hat{z}_{i,t}$, which are predicted by observed $s_{i,t}$ and estimated parameters $\hat{\Gamma}$. Also note that values of $z_{i,t}$ are residualized from individual and time-fixed effects. Therefore, the R^2 here can be thought of as the fraction of the variations in macroeconomic beliefs net of common and individual-specific factors that can be purely explained by partisan factors.

D Data Appendix

This online appendix provides the specific questions of the variables we use for our empirical analysis.

D.1 Michigan Survey of Consumers

Variable Description	Question Number	Question Text
Prob of higher unemployment rate	Q11	How about people out of work during the coming 12 months – do you think that there will be more unemployment than now, about the same, or less unemployment than now?
Prob of losing job	Q27b	During the next five years, what do you think the chances are that you (or your spouse/partner) will lose a job that you wanted to keep?
Prob of higher nominal income	Q16a	By about what percent do you expect your total (family) income to increase/decrease during the next 12 months?
Higher or lower real household income	Q15	During the next year or two, do you expect that your (family) income will go up more than prices will go up, about the same as prices go up, or less than prices will go up?
Better or worse personal financial status (future)	Q3	Now looking ahead—do you think that a year from now you (and your family living there) will be better off financially, worse off financially, or just about the same as now?
Better or worse personal financial status (past)	Q1	We are interested in how people are getting along financially these days. Would you say that you (and your family living there) are better off or worse off financially than you were a year ago?
Expected Increase in gas prices 1y	Q21a	About how many cents per gallon do you think gasoline prices will go up/down during the next five years compared to now?
Expected Increase in gas prices 5y	Q22a	About how many cents per gallon do you think gasoline prices will go up/down during the next twelve months compared to now?
Expected inflation rate	Q13a	By about what percent do you expect prices to go up/down on the average, during the next 12 months?
Expected inflation rate	Q14a	By about what percent per year do you expect prices to go up/down on the average, during the next 5 to 10 years?
Interest rates rise/fall	Q12	No one can say for sure, but what do you think will happen to interest rates for borrowing money during the next 12 months –will they go up, stay the same, or go down?
Business conditions improve/decline	Q8	And how about a year from now, do you expect that in the country as a whole, business conditions will be better or worse than they are at present, or just about the same?
Business financial conditions improve/decline 1y	Q5	Now turning to business conditions in the country as a whole—do you think that during the next 12 months we'll have good times financially or bad times financially?
Business financial conditions improve/decline 5y	Q9	Looking ahead, which would you say is more likely – that in the country as a whole we'll have continuous good times during the next 5 years or so, or that we will have periods of widespread unemployment or depression?
Index of Consumer Sentiment		Q1 + Q3 + Q5 + Q9 + Q19
Index of Current Economic Conditions		Q1 + Q19

Index of Consumer Expectations	$Q3 + Q5 + Q9$
Macro Sentiment Index	$Q5 + Q9$
Micro Sentiment Index	$Q1 + Q3 + Q19$

Table A.26: Data Description: Michigan Survey of Consumers

D.2 NY Fed Survey of Consumer Expectations

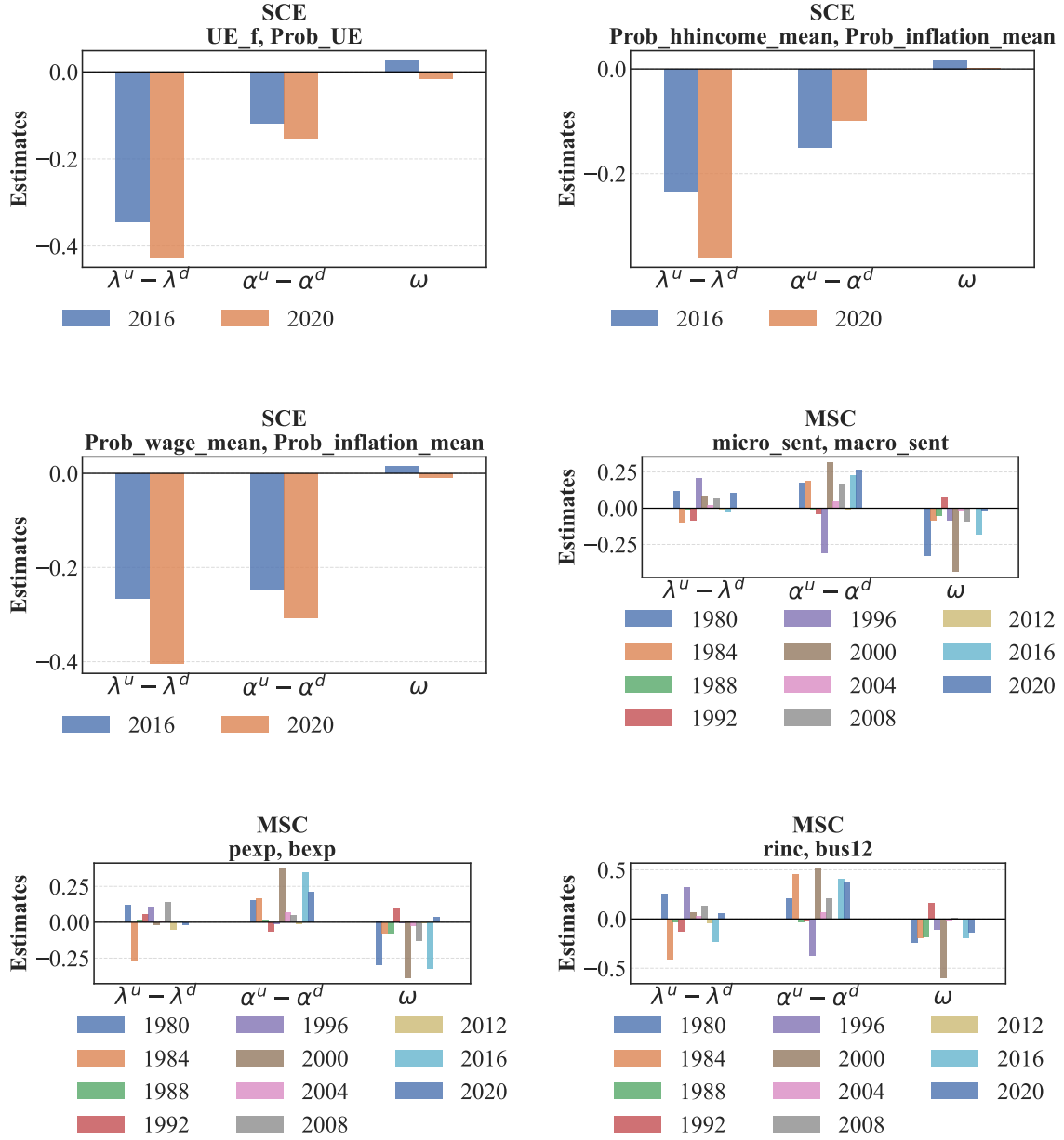
Variable Description	Question Number	Question Text
Prob of higher unemployment rate	Q4new	What do you think is the percent chance that 12 months from now the unemployment rate in the U.S. will be higher than it is now?
Perceived job separation prob	Q13new	What do you think is the percent chance that you will lose your ["main" if Q11>1, "current" if Q11=1] job during the next 12 months?
Perceived job finding prob	Q22new	Suppose you were to lose your ["main" if Q11>1] job this month. What do you think is the percent chance that within the following 3 months, you will find a job that you will accept, considering the pay and type of work?
Prob of quitting	Q14new	What do you think is the percent chance that you will leave your ["main" if Q11>1, "current" if Q11=1] job voluntarily during the next 12 months?
Higher or lower personal wage rate	Q23v2	Please think ahead to 12 months from now. Suppose that you are working in the exact same ["main" if Q11>1] job at the same place you currently work, and working the exact same number of hours. What do you expect to have happened to your earnings on this job, before taxes and deductions?
Percentage of real wage growth	Q23v2part2	By about what percent do you expect your earnings to have [increased/decreased as in Q23]? Please give your best guess.
Higher or lower household income	Q25v2	Over the next 12 months, what do you expect will happen to the total income of all members of your household (including you), from all sources before taxes and deductions?
Percentage of real household income growth	Q25v2part2	By about what percent do you expect your total household income to [increase/decrease as in Q25v2]? Please give your best guess.
Higher or lower spending	Q26v2	Over the next 12 months, what do you expect will happen to the total spending of all members of your household (including you)?
Percentage of spending growth	Q26v2part2	By about what percent do you expect your total household spending to [increase/decrease as in Q26v2]? Please give your best guess.
Expected Increase in gas prices	C4.1	Twelve months from now, what do you think will have happened to the price of the following items? Price of a gallon of oil
Expected Increase in food prices	C4.2	Twelve months from now, what do you think will have happened to the price of the following items? Price of food

Expected Increase in medical prices	C4_3	Twelve months from now, what do you think will have happened to the price of the following items? Price of medical care
Expected increase in college prices	C4_4	Twelve months from now, what do you think will have happened to the price of the following items? Cost of a college education
Expected increase in rent	C4_5	Twelve months from now, what do you think will have happened to the price of the following items? Cost of renting a typical house/apartment
Expected increase in gold	C4_6	Twelve months from now, what do you think will have happened to the price of the following items? Price of gold
Inflation or deflation	Q8v2	The next few questions are about inflation. Over the next 12 months, do you think that there will be inflation or deflation?
Expected inflation rate	Q8v2part2	What do you expect the rate of [inflation (if Q8v2=inflation)/deflation (if Q8v2=deflation)] to be over the next 12 months? Please give your best guess.
Higher or lower house prices	Q31v2	Over the next 12 months, what do you expect will happen to the average home price nationwide?
Expected house price change	Q31v2part2	By about what percent do you expect the average home price to [increase/decrease as in Q31v2]? Please give your best guess.
Prob of higher interest rate	Q5new	What do you think is the percent chance that 12 months from now the average interest rate on saving accounts will be higher than it is now?
Prob of higher stock market	Q6new	What do you think is the percent chance that 12 months from now, on average, stock prices in the U.S. stock market will be higher than they are now?
Federal welfare benefits: direction	qp1x5	What do you think is the percent chance that over the next 12 months there will be an increase/expansion, decrease/reduction, or no change in each of the following? Federal welfare benefits
Unemployment benefits: direction	qp1x6	What do you think is the percent chance that over the next 12 months there will be an increase/expansion, decrease/reduction, or no change in each of the following? Unemployment benefits
Payroll tax rate: direction	qp1x9	What do you think is the percent chance that over the next 12 months there will be an increase/expansion, decrease/reduction, or no change in each of the following? Payroll tax rate
Capital gains tax rate: direction	qp1x11	What do you think is the percent chance that over the next 12 months there will be an increase/expansion, decrease/reduction, or no change in each of the following? Capital gains tax rate
Average income tax rate: direction	qp1x16	What do you think is the percent chance that over the next 12 months there will be an increase/expansion, decrease/reduction, or no change in each of the following? Average income tax rate
Income tax rate for highest income bracket: direction	qp1x17	What do you think is the percent chance that over the next 12 months there will be an increase/expansion, decrease/reduction, or no change in each of the following? Income tax rate for highest income bracket
Federal welfare benefits: effect	qp2x5	If the following policy changes were implemented in the next 12 months, how would you and your household be impacted? Federal welfare benefits

Unemployment benefits: effect	qp2x6	If the following policy changes were implemented in the next 12 months, how would you and your household be impacted? Unemployment benefits
Payroll tax rate: effect	qp2x9	If the following policy changes were implemented in the next 12 months, how would you and your household be impacted? Payroll tax rate
Capital gains tax rate: effect	qp2x11	If the following policy changes were implemented in the next 12 months, how would you and your household be impacted? Capital gains tax rate
Average income tax rate: effect	qp2x16	If the following policy changes were implemented in the next 12 months, how would you and your household be impacted? Average income tax rate
Income tax rate for highest income bracket: effect	qp2x17	If the following policy changes were implemented in the next 12 months, how would you and your household be impacted? Income tax rate for highest income bracket

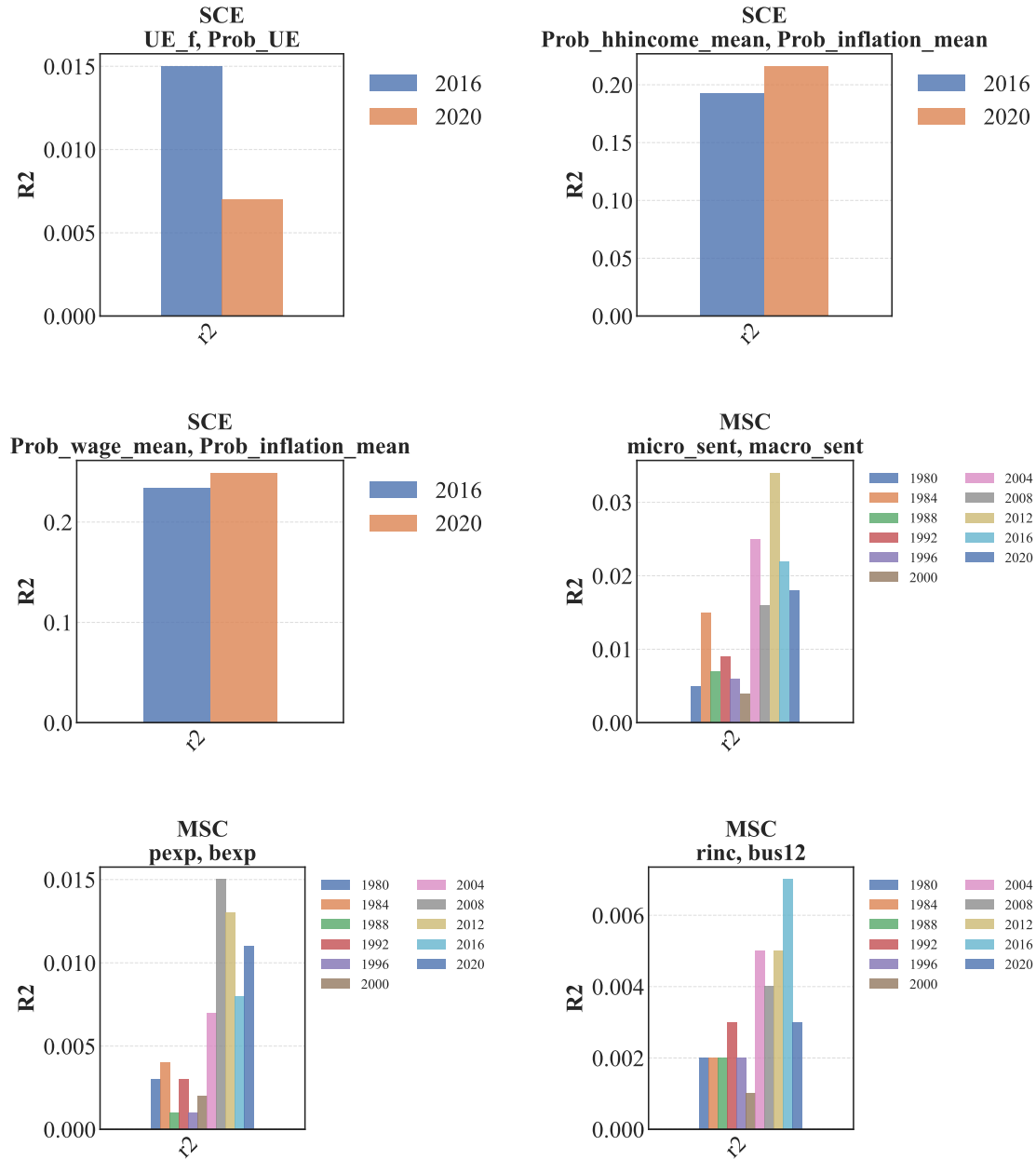
Table A.27: Data Description: Survey of Consumer Expectations

Figure A.4: Model Estimates



These figures plot the parameter estimates for each survey, each pair of micro and macroeconomic expectations, and each election cycle. Superscripts u and d represent *up* and *down* for brevity.

Figure A.5: Time-varying Importance of Partisanship in Macroeconomic Expectations Implied by Model Estimates



Notes: The figure plots the R^2 of a linear regression of observed cross-sectional macroeconomic expectations on their predicted values from the model estimates, for each pair of micro and macroeconomic beliefs, and each election cycle.

Table A.25: Model Estimates

			slope_up	slope_down	shift_up	shift_down	bias	coeff	r2	p_value
pexp	bexp	dem 1980	0.301	0.178	0.180	0.027	-0.298	0.055	0.003	0.000
		rep 1980	0.309	0.167	-0.285	-0.084	0.404	0.054	0.003	0.000
		dem 1984	0.129	0.394	0.094	-0.074	-0.074	0.058	0.004	0.000
		rep 1984	0.361	0.126	-0.127	0.052	0.052	0.052	0.003	0.000
		dem 1988	0.202	0.188	0.030	0.015	-0.077	0.032	0.001	0.000
		rep 1988	0.206	0.184	0.039	-0.006	-0.045	0.025	0.001	0.001
		dem 1992	0.277	0.220	-0.071	-0.010	0.096	0.057	0.003	0.000
		rep 1992	0.352	0.154	0.126	-0.050	-0.074	0.054	0.003	0.000
		dem 1996	0.261	0.152	-0.003	0.008	-0.003	0.033	0.001	0.000
		rep 1996	0.246	0.194	0.023	-0.021	0.023	0.042	0.002	0.000
		dem 2000	0.211	0.226	0.361	-0.013	-0.389	0.045	0.002	0.000
		rep 2000	0.293	0.149	-0.311	0.001	0.368	0.054	0.003	0.000
		dem 2004	0.300	0.301	0.039	-0.026	-0.026	0.079	0.007	0.000
		rep 2004	0.288	0.313	-0.057	0.026	0.026	0.090	0.009	0.000
		dem 2008	0.420	0.282	0.100	0.051	-0.128	0.114	0.015	0.000
		rep 2008	0.281	0.421	-0.065	-0.018	0.118	0.117	0.015	0.000
		dem 2012	0.329	0.377	-0.003	0.008	-0.003	0.106	0.013	0.000
		rep 2012	0.398	0.319	0.009	-0.011	0.009	0.117	0.016	0.000
		dem 2016	0.300	0.299	0.324	-0.025	-0.323	0.084	0.008	0.000
		rep 2016	0.337	0.270	-0.287	0.013	0.339	0.085	0.008	0.000
		dem 2020	0.340	0.355	0.089	-0.119	0.033	0.101	0.011	0.000
		rep 2020	0.354	0.345	-0.068	0.050	0.027	0.118	0.016	0.000
MSC	rinc	dem 1980	0.301	0.048	0.193	-0.021	-0.242	0.038	0.002	0.000
		rep 1980	0.047	0.285	-0.389	-0.002	0.417	0.026	0.001	0.000
		dem 1984	0.078	0.487	0.253	-0.198	-0.198	0.037	0.002	0.000
		rep 1984	0.453	0.062	-0.330	0.135	0.135	0.063	0.005	0.000
		dem 1988	0.189	0.225	0.040	0.071	-0.188	0.043	0.002	0.000
		rep 1988	0.158	0.253	-0.017	-0.089	0.145	0.046	0.002	0.000
		dem 1992	0.137	0.260	-0.066	-0.056	0.159	0.049	0.003	0.000
		rep 1992	0.246	0.169	0.120	0.088	-0.249	0.042	0.002	0.000
		dem 1996	0.343	0.016	-0.113	0.261	-0.113	0.040	0.002	0.000
		rep 1996	0.040	0.260	0.127	-0.114	0.127	0.042	0.002	0.000
		dem 2000	0.196	0.130	0.527	0.014	-0.601	0.032	0.001	0.000
		rep 2000	0.196	0.133	-0.490	0.013	0.557	0.019	0.000	0.013
		dem 2004	0.299	0.273	0.040	-0.027	-0.027	0.070	0.005	0.000
		rep 2004	0.261	0.312	-0.033	0.015	0.015	0.081	0.007	0.000
		dem 2008	0.292	0.162	0.098	-0.116	0.014	0.060	0.004	0.000
		rep 2008	0.186	0.268	-0.075	0.085	-0.016	0.047	0.002	0.000
		dem 2012	0.264	0.303	-0.001	0.002	-0.001	0.069	0.005	0.000
		rep 2012	0.309	0.264	-0.000	0.001	-0.000	0.078	0.007	0.000
		dem 2016	0.141	0.375	0.297	-0.111	-0.197	0.076	0.007	0.000
		rep 2016	0.346	0.174	-0.248	0.130	0.152	0.073	0.006	0.000
		dem 2020	0.271	0.211	0.255	-0.126	-0.133	0.049	0.003	0.000
		rep 2020	0.275	0.213	-0.173	0.017	0.236	0.046	0.002	0.000
micro_sent	macro_sent	dem 1980	0.302	0.184	0.203	0.028	-0.330	0.068	0.005	0.000
		rep 1980	0.221	0.258	-0.351	-0.106	0.496	0.066	0.005	0.000
		dem 1984	0.325	0.424	0.107	-0.085	-0.085	0.113	0.015	0.000
		rep 1984	0.419	0.314	-0.155	0.063	0.063	0.120	0.016	0.000
		dem 1988	0.287	0.298	0.008	0.024	-0.056	0.082	0.007	0.000
		rep 1988	0.276	0.305	-0.087	-0.118	0.281	0.091	0.009	0.000
		dem 1992	0.247	0.335	-0.053	-0.013	0.081	0.088	0.009	0.000
		rep 1992	0.278	0.309	0.033	0.012	-0.054	0.085	0.008	0.000
		dem 1996	0.375	0.169	-0.089	0.221	-0.089	0.075	0.006	0.000
		rep 1996	0.144	0.344	0.145	-0.122	0.145	0.086	0.008	0.000
		dem 2000	0.287	0.203	0.362	0.046	-0.440	0.060	0.004	0.000
		rep 2000	0.256	0.237	-0.311	-0.020	0.412	0.063	0.004	0.000
		dem 2004	0.410	0.389	0.030	-0.020	-0.020	0.148	0.025	0.000
		rep 2004	0.385	0.416	-0.040	0.018	0.018	0.133	0.020	0.000
		dem 2008	0.386	0.321	0.140	-0.029	-0.095	0.118	0.016	0.000
		rep 2008	0.332	0.374	-0.109	0.034	0.105	0.113	0.014	0.000
		dem 2012	0.430	0.439	-0.002	0.005	-0.002	0.168	0.034	0.000
		rep 2012	0.445	0.426	0.004	-0.007	0.004	0.162	0.031	0.000
		dem 2016	0.377	0.404	0.199	-0.031	-0.182	0.136	0.022	0.000
		rep 2016	0.411	0.373	-0.185	0.042	0.179	0.142	0.024	0.000
		dem 2020	0.432	0.324	0.144	-0.122	-0.024	0.123	0.018	0.000
		rep 2020	0.432	0.331	-0.104	-0.004	0.167	0.121	0.017	0.000
SCE	UE_s	dem 2016	-0.048	0.156	0.415	0.117	0.002	0.134	0.016	0.000
		rep 2016	-0.035	0.145	0.449	0.083	-0.010	0.139	0.017	0.000
		dem 2020	-0.111	0.224	0.495	-0.039	-0.059	0.070	0.005	0.000
		rep 2020	-0.096	0.206	0.441	-0.007	-0.053	0.079	0.006	0.000
	UE_f	dem 2016	-0.180	0.165	0.260	0.379	0.026	0.135	0.016	0.000
		rep 2016	-0.163	0.148	0.310	0.338	0.015	0.133	0.016	0.000
		dem 2020	-0.245	0.181	0.193	0.348	-0.015	0.081	0.006	0.000
		rep 2020	-0.217	0.164	0.212	0.347	-0.043	0.073	0.005	0.000
	Prob_wage_mean	dem 2016	-0.077	0.189	-0.960	-0.715	0.014	0.486	0.229	0.000
		rep 2016	-0.056	0.169	-0.943	-0.731	0.015	0.492	0.229	0.000
		dem 2020	-0.132	0.272	-1.025	-0.718	-0.008	0.511	0.253	0.000
		rep 2020	-0.071	0.154	-0.811	-0.964	-0.024	0.513	0.254	0.000
	Prob_hhincome_mean	dem 2016	-0.095	0.141	-0.950	-0.800	0.016	0.431	0.187	0.000
		rep 2016	-0.042	0.088	-0.905	-0.849	0.022	0.438	0.190	0.000
		dem 2020	-0.158	0.202	-0.971	-0.873	0.001	0.459	0.216	0.000
		rep 2020	-0.086	0.122	-0.794	-1.020	-0.041	0.464	0.221	0.000
	Prob_wage	dem 2016	-0.085	0.052	-0.028	0.396	-0.011	0.065	0.004	0.000
		rep 2016	-0.047	0.019	0.056	0.336	-0.019	0.064	0.004	0.000
		dem 2020	-0.115	0.080	-0.076	0.456	-0.068	0.038	0.001	0.000
		rep 2020	0.064	-0.105	0.313	-0.061	0.072	0.042	0.002	0.000